

Anthesis: An Agentic Software Development Lifecycle

Anthesis Whitepaper (Draft)

Versioning & Authors

Version	Date	Author	Notes
0.1.0	2025-12-10	Ryan Jennings	Initial draft compiled from Meristem RFCs.
0.1.1	2026-01-18	Ryan Jennings	Expanded details and organizing
0.1.2	2026-02-12	Ryan Jennings	Patch update to whitepaper overview.
0.1.3	2026-02-22	Ryan Jennings	Terminology and source-attribution drift fixes.
0.1.4	2026-04-21	Ryan Jennings	Listed the current Phloem MCP tool surface explicitly.
0.2.0	2026-05-20	Ryan Jennings	Updated for recent governance and budget-control changes.

1. Executive Summary

Anthesis is a governed execution state machine for agentic SDLC, designed to enable autonomous software delivery without surrendering human authority, auditability, or safety.

It treats governance as a first-class system boundary. Autonomy is permitted only when context is complete, risk is understood, and approvals are explicit.

Primary sources:

- CHARTER.md
- meristem/rfc_0001_agentic_sdlc_platform.md

2. System Overview

Anthesis centers on the Model Context Protocol (MCP) as the orchestration and governance layer. MCP coordinates events, agent execution, and audit trails while preserving Git as the system of record.

This separation allows automation to operate safely: agents never mutate Git directly, and all actions flow through MCP-controlled proposals, approvals, and state transitions.

Current MCP Tool Surface Phloem currently exposes these MCP tools through the Anthesis MCP runtime:

Tool	Purpose
system_status	Read current MCP/system status from Phloem.
create_event	Create a governed event in Phloem.
get_event	Retrieve an event by id.
run_agent	Start an agent execution for an existing event.
invoke_agent	Atomically create an event and start an agent execution.
list_executions	List executions, optionally filtered by repository, execution state, state, or limit.
get_execution	Retrieve execution status and details by id.
get_execution_logs	Retrieve logs for an execution.
list_proposals	List proposals, optionally filtered by repository.
get_proposal	Retrieve a proposal by id.
decide_proposal	Approve or reject a proposal with a recorded reason.
list_approvals	List approvals, optionally filtered by proposal, execution, or repository.
audit_logs	Fetch aggregate audit logs, optionally filtered by execution, proposal, or repository.
risk_summary	Read the current risk-tier summary.

Tool	Purpose
<code>list_tools</code>	List MCP-exposed Phloem tools from the tool registry endpoint.

The `/v1/tools` registry endpoint currently also advertises `hello_ai` as a sanity-check tool.

Primary sources:

- `meristem/rfc_0001_agentic_sdlc_platform.md`
- `meristem/rfc_0002_mcp_orchestration_api.md`
- `services/phloem/mcp_runtime.py`
- `services/phloem/routes/tools.py`

3. Governance and Safety Model

Anthesis enforces governance via Calyx policies, risk tiers, and a formal state machine. Approvals gate risky actions and ensure human oversight.

Policy evaluation and state transitions are auditable. This makes the system deterministic about when an agent can act, when it must pause, and how it recovers from failure.

Recent governance work broadens this model from approval gates alone toward governed execution criteria. Anthesis now treats execution transitions, replay evidence, temporal interaction context, recursive coordination, prompt profiles, model routing, and token/context budgets as policy-governed resources. The goal is not merely to permit or block an agent action, but to make the basis for that action reviewable, reproducible, and bounded.

Primary sources:

- `meristem/rfc_0003_approval_policy_engine.md`
- `meristem/rfc_0014_risk_classification_safety_tiers.md`
- `meristem/rfc_0020_state_machine_formal_specification.md`

4. Agent Model

Agents are defined by explicit schemas and execution contracts. They produce proposals and never self-approve. MCP enforces lifecycle rules and idempotency.

Agents are capability-scoped and operate under least-privilege constraints. Execution modes and retries are standardized to ensure predictable behavior across the SDL.

Primary sources:

- `meristem/rfc_0005_agent_schema.md`
- `meristem/rfc_0009_agent_execution_semantics.md`

5. Context and Determinism

Context assembly is deterministic, provenance-aware, and token-budgeted. This ensures reproducibility across agent runs.

Inputs are structured and attributable, preserving provenance from source artifacts, prompts, and event metadata. The result is a reproducible context snapshot for audits and replay.

Token budgeting is a governance control, not just a cost or performance optimization. A budget profile constrains input tokens, output tokens, preferred context sources, fallback expansion rules, summarization depth, model routing, and escalation behavior. For repository-scale review, Anthesis can require agents to consult routing indexes and summaries first, expand into normative source documents only when justified, and emit a budget ledger that records included, summarized, excluded, deferred, and escalated context.

This prevents large-context execution from becoming an invisible authority expansion. When an agent cannot complete a task within the approved budget, it must return a bounded result with explicit limitations or request approval to escalate to a larger context or model profile.

Primary sources:

- `meristem/rfc_0007_context_assembly_prompt_composition.md`
- `meristem/rfc_0004_provenance_ingestion.md`

6. Model Runtime Abstraction

Anthesis abstracts model runtimes so agents declare requirements rather than specific backends. MCP resolves runtimes safely and audits drift.

This approach supports local-first deployments while allowing controlled model evolution. Drift budgets and benchmarking constrain unexpected behavior changes.

Model routing is part of governance. Runtime selection should be explainable in terms of task class, budget profile, sensitivity, evidence requirements, and approved escalation path rather than hidden provider preference.

Primary sources:

- `meristem/rfc_0008_model_runtime_abstraction.md`
- `meristem/rfc_0025_model_evaluation_benchmarking_drift_budgets.md`

7. Observability and Evidence

Executions are fully observable with logs, traces, and audit trails. Evidence remains attributable and immutable.

Observability ties together events, executions, approvals, and outputs so incidents can be investigated and compliance evidence can be exported without ambiguity.

Replay evidence includes more than final outputs. It should capture execution identity, authority linkage, normalized environment context, activity references, outcome summaries, known nondeterminism, and budget/context decisions. This lets reviewers distinguish between a reproducible intervention, a bounded replay claim, and a degraded execution that requires additional review.

Primary sources:

- `meristem/rfc_0011_observability_telemetry_drift.md`
- `meristem/rfc_0023_audit_compliance_evidence_export.md`

8. Orchestration and Events

Repository events and explicit triggers initiate agent workflows. MCP schedules, retries, and coordinates execution.

Event ingestion provides a consistent entry point for automation, while execution controls prevent duplicate runs and enforce idempotency.

Primary sources:

- `meristem/rfc_0006_repository_event_trigger_model.md`
- `meristem/rfc_0009_agent_execution_semantics.md`

9. Interfaces and Control Surfaces

CLI, UI, and chat interfaces provide operator control. All interfaces map to MCP actions and enforce the same governance rules.

Human control surfaces are designed for transparency and recovery, including approvals, overrides, and kill switches, without granting hidden capabilities.

Primary sources:

- `meristem/rfc_0015_human_interfaces_control_surfaces.md`
- `meristem/rfc_0021_cli_ux_command_semantics.md`

10. Extensibility and Plugins

Plugins extend capability without expanding authority. Governance, auditability, and isolation remain mandatory.

Extensions are constrained by explicit contracts and policy enforcement so they cannot bypass MCP or elevate privileges.

Primary sources:

- `meristem/rfc_0026_plugin_extension_model.md`

11. Security and Identity

Secrets, credentials, and integration boundaries are explicitly managed. Trust boundaries are documented and enforced.

Agents operate without direct access to sensitive material unless explicitly authorized, and integration tokens are scoped and auditable.

Primary sources:

- `meristem/rfc_0012_secrets_credential_management.md`
- `meristem/rfc_0022_external_integrations_mcp_federation.md`

12. Lifecycle and Governance

RFCs, policies, agents, and models follow an explicit lifecycle and ownership model. Amendments are governed and auditable.

This lifecycle prevents silent drift, preserves intent, and ensures future agents can reason about historical decisions.

Primary sources:

- `meristem/rfc_0017_governance_ownership_rfc_lifecycle.md`
- `meristem/rfc_0099_v1_unfreeze_amendment_rules.md`

Glossary

- **Anthesis:** The governed execution state machine for agentic SDLC.
- **Anthesis (execution state):** The state in which agents are permitted to execute under governance controls.
- **Phloem (MCP):** The Model Context Protocol server that orchestrates events, context, policies, and execution.
- **Calyx:** Policy engine that evaluates approvals, risk tiers, and governance rules.
- **Bloom Class:** Risk tier classification used to gate approvals and execution.
- **Xylem:** Relay/worker execution plane and scheduling fabric.
- **Meristem:** Constitutional layer containing normative RFCs.
- **Inflorescence:** Multi-agent coordination graphs and workflow orchestration.
- **Budget Profile:** A reusable policy profile that constrains token usage, retrieval scope, summarization depth, model routing, and escalation behavior for an agent workflow.
- **Budget Ledger:** Audit evidence describing the approved budget, consumed tokens, context included or excluded, summaries generated, model routes used, and any escalation decisions.

Anthesis — Executive Summary

Anthesis enables governed, asynchronous background agents to automate every stage of the software development lifecycle, while governance ensures humans remain accountable, decisions auditable, and risk explicitly controlled.

The Problem Anthesis Addresses

AI tools can now:

- Generate code
- Modify repositories
- Execute tools
- Propose architectural changes

But they lack:

- Accountability
- Provenance
- Clear authority boundaries
- Auditable decision trails

In most environments today, AI is either:

- **Over-trusted** (silent changes, auto-execution), or
- **Under-used** (manual copy/paste, no integration)

Anthesis exists to answer:

How can AI assist real software development without undermining safety, governance, or trust?

What Anthesis Is

Anthesis is:

- A **governance system** for AI-assisted SDLC
- A **policy-driven control plane** for agents and tools
- A **Git-native source of truth** for intent, decisions, and state
- A **human-in-the-loop framework** by design

It treats AI agents as **contributors**, not authorities.

What Anthesis Is Not

Anthesis is **not**:

- A code generator
- A replacement for IDE tools like Copilot
- A single AI model or vendor platform
- An autonomous “auto-ship” system

Anthesis governs *how* those tools may be used — not *which* tools you use.

Core Principles

- 1. Git Is the Source of Truth** All plans, decisions, approvals, and outcomes are recorded in version control.
 - 2. Agents Propose — Humans Approve** Agents may analyze, draft, and recommend. Final authority remains human.
 - 3. Everything Is Auditable** Every action has provenance: who or what proposed it, under which policy, and why.
 - 4. Risk Determines Oversight** Higher-risk changes require stronger review and approval guarantees.
 - 5. No Silent Execution** No tool execution or state mutation occurs without an explicit, recorded transition.
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How Anthesis Works (Conceptual Flow)

- 1. Intent is declared** A task, change, or investigation is proposed.
 - 2. Agents assist** AI agents analyze context, generate plans, draft artifacts, or surface risks.
 - 3. Policies evaluate risk** Governance rules determine required approvals and constraints.
 - 4. Humans review and approve** Decisions are explicit and recorded.
 - 5. Execution is controlled** Tools run only within approved boundaries.
 - 6. State is recorded** Outcomes and rationale are committed to Git.
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Why This Matters (Especially for Regulated Environments)

Anthesis enables:

- Safe adoption of AI in production SDLCs

- Clear audit trails for compliance and incident review
- Controlled use of external tools and models
- Incremental rollout with increasing governance

It is designed for environments where:

- Security matters
 - Compliance matters
 - Mistakes are costly
 - “The agent did it” is not an acceptable explanation
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The Role of RFCs

Anthesis is governed by a set of **living RFCs**.

- RFCs define what is allowed, forbidden, and required
- RFCs are precise, versioned, and auditable
- This document explains *why* Anthesis exists
- The RFCs define *how Anthesis works*

If there is a conflict, RFCs always win.

Adoption Model

Anthesis is designed to be adopted:

- Gradually
- Incrementally
- Alongside existing tools and workflows

Organizations can start with:

- Planning and review governance
- Then expand to controlled execution
- Then introduce more advanced agent participation

Anthesis Governed Agentic SDLC Platform PRD

Anthesis is a governed agentic SDLC orchestration platform purpose-built to coordinate AI agents against code repositories while enforcing strict human-in-the-loop controls, policy gates, and comprehensive audit trails. The platform enables automation with deterministic, safe, and fully traceable execution: teams can scale agent workflows without sacrificing control, governance, or risk tolerance. Beyond blocking unsafe changes, Anthesis converts observed governance gaps into reviewed, reusable policy criteria so the system becomes safer from real evidence over time.

Goals

Business Goals

- Achieve 90% reduction in ungoverned code changes by enforcing mandatory policy gates and human approvals.
- Enable safe deployment of AI agent workflows in at least 5 enterprise SDLC environments within the first year.
- Demonstrate regulatory and audit compliance for AI-powered SDLC pipelines, opening up new enterprise sales opportunities.
- Drive 50% adoption among engineering leads and platform engineers within pilot client organizations.
- Decrease mean-time-to-remediate risky agent behaviors/events below 2 hours via streamlined oversight and rollback tools.

User Goals

- Ensure all agent-driven code changes are subject to explicit, auditable human approval or rejection.
- Empower platform engineers and ICs to safely compose, propose, and manage complex agent workflows without risk of policy violation.
- Give engineering leads full, real-time visibility and control over agentic SDLC processes, preventing accidental or unauthorized changes.
- Streamline the agent workflow proposal/approval lifecycle to avoid friction while maintaining oversight.
- Provide clear, machine-and-human readable records of all agent actions and governance events.

Non-Goals

- Anthesis does not aim to replace CI/CD pipeline tools; it orchestrates agentic workflows and integrates with existing pipelines.
- The platform will not operate in unsupervised or autonomous mode; human-in-the-loop is always required for sensitive or production-impacting actions.
- Anthesis is not an AI agent development platform; it serves as the orchestrator and policy/governance layer for existing agent workflows.
- Anthesis does not silently self-modify governance policy; learned criteria must be reviewed and approved before becoming active controls.

User Stories

Personas 1. Engineering Lead

- As an Engineering Lead, I want to review and approve AI agent proposals before execution, so that I can ensure all actions comply with company policy and best practices.
- As an Engineering Lead, I want a complete audit trail of agent activity and approvals, so that I can quickly address compliance requests.

2. Platform Engineer

- As a Platform Engineer, I want to define, classify, and compose agentic workflows using policy templates, so that deterministic and compliant automation is guaranteed.
- As a Platform Engineer, I want to trigger proposals via CLI and assign reviewers based on required clearance levels, so that appropriate oversight is enforced.
- As a Platform Engineer, I want escaped defects to be converted into proposed policy criteria, so that governance improves from real evidence rather than static rules alone.
- As a Platform Engineer, I want to define token and context budget profiles for agent workflows, so that cost, context exposure, retrieval scope, and model routing are governed before execution begins.

3. Individual Contributor (IC)

- As an IC, I want to submit code or infrastructure change proposals through Anthesis, so that my work benefits from automated agent support while staying within governance.
- As an IC, I want machine-readable feedback on rejected proposals, so that I can rapidly iterate and comply with required revisions.
- As an IC, I want to see which agent actions require human approval, so that I can track and accelerate the review process.

Functional Requirements

- **Orchestration Core (Phloem) — Priority: Critical**
 - *Agent Workflow Orchestration:* Execute, monitor, and manage AI agent-driven SDLC workflows.

- *State Machine Enforcement*: Enforce allowable transitions, preventing agents from acting outside approved workflow states.
- *Output Standardization*: Normalize agent outputs for deterministic audit and review.
- **Governance & Controls — Priority: Critical**
 - *Policy Definition & Enforcement*: Allow admins to create/tune policy gates, clearance classifications, and risk thresholds.
 - *Approval Workflow*: Route agent proposals through multi-stage human approval with granular, role-based access controls.
 - *Audit Logging & Traceability*: Record every agent and human interaction/decision in an immutable, queryable audit log.
 - *Feedback-to-Policy Lifecycle*: Capture missed violations, route proposed new criteria through human review, and version approved controls for future evaluations.
 - *Token and Context Budget Governance*: Enforce policy-defined budgets for input tokens, output tokens, retrieval scope, summarization depth, model routing, and escalation behavior before and during agent execution.
- **User Interface & CLI — Priority: High**
 - *CLI Submission*: Allow users to compose, test, and submit proposals directly from the command line.
 - *Feedback & Notification*: Deliver immediate, clear CLI or webhook-based feedback for approvals, rejections, and policy violations.
 - *Machine/Human-Readable Outputs*: Output workflow status, proposal metadata, and audit records in formats suitable for both human and machine consumption.
- **Policy/Classification Library — Priority: Medium**
 - *Reusable Policy Templates*: Catalog of company/vetting policies, risk classifications, and action boundaries for workflow reuse.
 - *Classification Levels*: Allow for automatic classification tags (e.g., LOW, MEDIUM, HIGH RISK) and clearance requirements on agent actions.
 - *Budget Profiles*: Catalog reusable budget profiles for common workflow classes, such as small code review, repository-scale RFC review, security-sensitive analysis, and approved escalation paths.
- **Oversight & Rollback Tools — Priority: Medium**

- *Rollbacks*: Manual or automated reversal of agent actions with provenance management.
 - *Approval/Denial Comments*: Require rationale for all human overrides/denials to strengthen audit and learning.
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User Experience

Entry Point & First-Time User Experience

- Users discover Anthesis either through internal onboarding documentation or via direct CLI/console prompts in their development environments.
- On first use, the CLI provides a guided onboarding flow: user authentication, selection of clearance level, introduction to policy gates, and a sample agentic workflow proposal.
- A tutorial (CLI/console-based) walks users through proposal composition, policy selection, and the review/approval cycle with stepwise feedback.

Core Experience

- **Step 1:** User composes an agentic workflow (e.g., refactor codebase using Agent X) and submits for proposal via Anthesis CLI.
 - Intuitive CLI command structure; prompts for required metadata and classification.
 - Validates proposal structure, flags missing fields, and prevents submission if out of policy.
- **Step 2:** Proposal appears in the governance dashboard (or mirrored CLI feed) for required human review.
 - Role-based access: Only reviewers with sufficient clearance see/rule on the item.
 - Real-time notification to reviewers (email, Slack, webhook).
- **Step 3:** Reviewer opens the proposal, sees policy diff, agent action summary, and risk classification.
 - Provides approve/deny buttons and optional comment box.
 - Record is written to audit log, including rationale if denied.
- **Step 4:** Upon approval, Anthesis executes the agent workflow; output is streamed back to the CLI in real time.
 - Execution is paused if new policy violations are detected post-approval.

- **Step 5:** Every state transition—proposal, approval, execution, post-execution classification—is both human- and machine-readable, (e.g., JSON, RFC-style outline).
 - Rollback option available immediately on post-execution events.

Advanced Features & Edge Cases

- Power users can save and share reusable workflow/policy templates within their organization.
- Supports bulk-approval or multi-proposal batch review for high-velocity pipelines.
- If an agent attempts an unauthorized transition, Anthesis halts execution and files an incident for escalation review.
- Handles disconnected/failed review scenarios by auto-expiring stale proposals and alerting the submitter.
- **Governed feedback loop example:** While building a React Native SDK, an agent accidentally introduces DOM-specific usage such as `react-dom`, `document`, or `window`. Anthesis should detect the mismatch from the proposed diff and surface it as a governance violation before acceptance. If the violation is not already covered by active policy, the escaped defect should feed back into the governance system as a new reusable criterion, such as “React Native SDK code must not depend on DOM APIs,” so future proposals are evaluated against the learned control.
- **Governed token budgeting example:** A developer asks Anthesis to review a pull request or draft RFC against a large repository of canonical RFCs, summaries, planning documents, source files, and prior review comments. Rather than allowing an agent to read arbitrarily broad context, Anthesis applies a policy-bound budget profile before execution begins. The profile constrains maximum input and output tokens, preferred context sources, fallback expansion rules, summarization depth, model routing, and escalation conditions. The agent must consult routing/index material first, expand into normative source documents only when justified, and produce a budget ledger showing what context was included, summarized, excluded, or deferred. If the task cannot be completed within the approved budget, the workflow must either return a bounded result with explicit limitations or request approval to escalate to a larger context/profile.

UI/UX Highlights

- CLI output is color-coded for warnings/errors and uses ANSI-safe formatting for accessibility.

- Responsive web dashboard (for audit/review) supports keyboard navigation and screen readers.
 - All outputs and records can be exported in compliance-friendly, immutable formats (YAML, JSON, RFC text).
 - Minimal multi-step flows; every action requires explicit confirmation and provides context for next steps.
 - High-contrast, scalable elements for both terminal and web UIs to meet AA accessibility standards.
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Narrative

Taylor, an engineering lead at a large fintech firm, worries about the risk and opacity of letting AI agents modify their codebase. Her team wishes to automate routine maintenance using modern agent tools, but CTO mandates require auditable, policy-governed pipelines—no exceptions.

With Anthesis, Taylor’s platform engineers hook into their code repositories and orchestrate workflows using Phloem. An individual contributor submits a refactoring task via CLI; the workflow is automatically classified as “Medium Risk” due to the codebase size, requiring explicit review. Taylor receives a detailed summary—policy diff, proposed changes, required clearance, and token/context budget profile—right in her Anthesis dashboard and can approve, reject, or escalate the execution budget with rationale.

Upon approval, Anthesis executes the workflow under strict state machine supervision, streaming every result, decision, model route, budget event, and agent output into a structured audit log. Weeks later, a client regulator requests a full trace: Taylor instantly exports an immutable, human- and machine-readable record of the entire event, including the evidence for which context was considered and why additional context was excluded.

By shifting from opaque automation to governed, agentic SDLC, Taylor not only accelerates the team’s work but dramatically reduces compliance overhead and mitigates operational risk—turning the AI revolution into a business advantage.

Success Metrics

User-Centric Metrics

- Number/proportion of agent proposals submitted through Anthesis vs. “shadow” automation workflows (target: >90% via Anthesis)
- User satisfaction score for core CLI/review UX (target: 4.5/5)
- Average review/approval cycle time (target: <30 minutes)

Business Metrics

- New enterprise customers acquired citing audit/governance as key driver (target: 5 in year one)
- Decrease in compliance incidents or risk events from agentic SDLC (target: 0 critical/major events in pilot phase)
- Percentage reduction in manual policy review workload for leads/managers (target: 60%)
- Percentage of confirmed escaped defects converted into reviewed policy criteria
- Median time from confirmed escaped defect to approved reusable control
- Percentage of agent executions completed within approved budget profile without unapproved escalation

Technical Metrics

- System uptime for orchestration engine and CLI (target: >99.9%)
- Median latency for workflow proposal to execution (<2 minutes under normal load)
- Audit log integrity (zero detected inconsistencies, tampering, or gaps)
- Repeat rate of previously observed violation classes after an approved reusable control is activated
- Budget ledger completeness for governed executions, including included, summarized, excluded, and escalated context records

Tracking Plan

- Agent workflow submission events
- Approval/denial actions and timings
- Policy violation detections and escalations
- Audit log access/export requests
- Rollback and incident/alert triggers
- Budget profile selection, token consumption, model routing, context inclusion/exclusion, summarization, and escalation events

Technical Considerations

Technical Needs

- **Orchestration engine (Phloem):** Manages agent workflow lifecycles, execution, and state transitions.
- **Policy engine:** Evaluates workflows against active policy templates/classifications at proposal and execution time.
- **CLI Client:** Cross-platform, secure, integrates with user auth and proposal APIs.
- **Data models:** Immutable audit log objects; proposal, workflow, approval, policy schema.
- **Budget ledger model:** Records approved token/context budget, model route, retrieval plan, summarization operations, context inclusions/exclusions, and escalation decisions.
- **Dashboard (optional in v1):** For review/approval traceability, policy management.
- **APIs:** Internal and (optionally) public endpoints for workflow submission, review, policy access.

Integration Points

- **SCM/code repository hooks** (e.g., GitHub, GitLab)
- **Notification/email/Slack/workflow integration** (for real-time approvals/reviews)
- **SIEM/Compliance tool export endpoints**

Data Storage & Privacy

- **Immutable audit/event log** stored in append-only, encrypted persistence (cloud or on-prem)
- **Separation of user data/proposal content** from minimal PII (all access/exports logged)
- **Meets SOC2, ISO, and regulator requirements** for SDLC traceability (for US/EU clients)

Scalability & Performance

- **Supports hundreds of concurrent workflow proposals/executions** per org (enterprise use case)
- **Designed for low-latency review/approval**, resilient to reviewer delays
- **Horizontal scaling** for orchestration and policy enforcement microservices

Potential Challenges

- Guaranteeing out-of-band agent activity is captured/prevented (requires deep SCM hooks and alerting)
- Handling workflows with overlapping/conflicting policy classifications
- Preventing budget optimization from degrading evidence quality, omitting canonical sources, or hiding uncertainty behind over-compressed summaries
- User education: ensuring leads/engineers understand the limits and powers of the agentic model
- Ensuring audit log/tamper protection in on-premise deployments

Anthesis SDLC Usage Example

This example shows how a team uses Anthesis inside a repository to automate the SDLC while keeping governance intact.

Phloem is the MCP server; references to MCP below refer to Phloem.

Scenario: Add a Billing Endpoint

- 1) An engineer opens a feature branch and updates `requirements/` with a short PRD note describing the billing endpoint.
- 2) A human runs:

```
anthesis agent run requirements "draft API requirements for billing endpoint"
```

MCP processes the run:

- Pre-Bloom: event recorded.
 - Context Assembly: RFCs and requirements pulled in.
 - Calyx: Bloom Class, artifact scope, and approval requirements evaluated.
 - Governance: conflict, signal, and interaction evidence checked before execution.
- 1) MCP creates a proposal and halts at `awaiting_approval`. An approver reviews and approves via CLI or UI.
 - 2) Phloem authorizes a scoped execution. The executor stages updated artifacts under `requirements/` or `meristem/` and records evidence.
 - 3) The engineer runs:

```
anthesis agent run implementation "implement billing endpoint"
```

- 1) MCP enforces policy, generates a proposal with code diffs, and waits for approval if required.
- 2) The approver approves the scoped artifacts; execution proceeds under Phloem control and records replay evidence.
- 3) The engineer runs:

```
anthesis agent run testing "add tests for billing endpoint"
```

- 1) MCP executes the testing agent and logs outcomes; CI runs as usual and is not bypassed.
- 2) Audit evidence is available via MCP (`/v1/audit/logs`) for compliance and review.

Outcome:

Every agent action is gated by Calyx, recorded via MCP, and replayable or auditable. Approvals are scoped to artifacts and execution candidates; humans remain in control.

Scenario: Repository Modification Triggers SDLC Workflow

This shows a repo-change-driven flow where the human does not call the CLI directly.

- 1) A developer updates `requirements/payment_api.md` and pushes a commit.
- 2) MCP receives the repo event (e.g., `git.commit`) via webhook or a workflow tool.
- 3) Calyx classifies changes under `requirements/**` as Bloom Class 2 and requires approval.
- 4) MCP assembles context and runs a requirements-review agent to identify gaps or draft updates.
- 5) MCP creates a proposal and halts at `awaiting_approval`.
- 6) An approver reviews and approves the scoped proposal; execution proceeds under policy and logs audit metadata.

Outcome:

The commit itself triggers SDLC automation, but all changes are still gated by policy and human approval.

Scenario: Deferred Execution with Manual Dispatch

This shows a queued execution that is explicitly dispatched later.

- 1) A developer runs:

```
anthesis agent run implementation "refactor payment adapter" --mode deferred
```

- 1) MCP creates the execution in `queued` lifecycle state.
- 2) An operator dispatches it when ready:

```
curl -X POST "$ANTHESIS_MCP_URL/v1/executions/<id>/dispatch" \
-H "X-Anthesis-Key: $ANTHESIS_API_KEY" \
-H "Content-Type: application/json" \
-d '{"reason":"manual dispatch","actor":"operator"}'
```

- 1) MCP advances through context, Calyx, conflict/signal checks, and either awaits approval or completes.

Scenario: Clarification via Approval Payload

This shows how to capture clarifying answers without mid-run input.

- 1) Trigger the agent run:

`anthesis agent run requirements "clarify billing rules"`

- 1) MCP creates a proposal that includes the question in **rationale**.
- 2) The approver supplies the answer in the approval reason (or attached payload via the UI).
- 3) MCP records the decision and continues execution only after the clarified intent is stabilized and policy permits it.

Documentation: Types and Conventions

- **RFCs** (Request for Comments) are proposals for changes or additions to the Anthesis system that may impact requirements and specifications. They provide a structured process for discussing and approving modifications to ensure alignment with the overall goals and design principles of Anthesis.
- **Architecture** documents the technical design and structure of the Anthesis system, including decision records, design notes, and diagrams.
- **Requirements** define the “what” - the necessary conditions and capabilities for the Anthesis system.
- **Specifications** define the “how” - the expected structure, behavior, and interfaces for components that fulfill the requirements.

Meristem: The Living RFCs of Anthesis

This directory is the public summary view of Anthesis Meristem RFCs.

- Each `rfc_*.md` file here is a concise, non-normative summary.
- The canonical, normative RFC source is `meristem/`.
- If any discrepancy exists between this directory and `meristem/`, `meristem/` is authoritative.
- This summary index is expected to mirror every canonical `meristem/rfc_*.md` file exactly once.
- The summary section in each public mirror file should match the canonical RFC summary from `meristem/` rather than introducing a divergent paraphrase.

Meristem is Anthesis’s constitutional layer: the RFC set that defines system behavior, governance, and constraints.

Included Public Summaries

- RFC-0000: Anthesis Terminology & Naming Convention
- RFC-0001: Agentic SDLC Platform
- RFC-0002: MCP Orchestration API
- RFC-0003: Approval Policy Engine
- RFC-0004: Provenance & Ingestion Model
- RFC-0005: Agent Schema
- RFC-0006: Repository Event & Trigger Model
- RFC-0007: Context Assembly & Prompt Composition
- RFC-0008: Model & Runtime Abstraction
- RFC-0009: Agent Execution Semantics
- RFC-0010: CI/CD Integration Model
- RFC-0011: Observability, Telemetry & Drift
- RFC-0012: Secrets & Credential Management
- RFC-0013: Multi-Agent Coordination & Chaining
- RFC-0014: Risk Classification & Bloom Classes
- RFC-0015: Human Interfaces & Control Surfaces
- RFC-0016: Dormancy & Disaster Recovery
- RFC-0017: Governance, Ownership & RFC Lifecycle
- RFC-0018: Anthesis Tooling & Project Bootstrap
- RFC-0019: Anthesis Core Distribution & Update Strategy
- RFC-0020: Anthesis State Machine (Formal Specification)
- RFC-0021: Anthesis CLI UX & Command Semantics
- RFC-0022: External Integrations & MCP Federation
- RFC-0023: Audit, Compliance & Evidence Export
- RFC-0024: Anthesis UI & Visualization Standards
- RFC-0025: Model Evaluation, Benchmarking & Drift Budgets
- RFC-0026: Anthesis Plugin & Extension Model
- RFC-0027: Configuration Model
- RFC-0028: Repo Registry & Identity Resolution
- RFC-0029: Policy Composition & Override Resolution
- RFC-0030: Approval-Artifact Mapping
- RFC-0031: Agent Classes & Trust Model
- RFC-0032: External Tool Integration & Admission
- RFC-0033: Proposal Taxonomy & Approval Scoping Strategy
- RFC-0034: Sessions & Prompt Context Governance
- RFC-0035: Operating Modes (Offline / Local / Remote)
- RFC-0036: QART Engineering
- RFC-0037: Drift Loop Engineering
- RFC-0038: Unified Governance Control Doctrine
- RFC-0039: MCP Transport Profiles (`http`, `stdio`, `sse`)
- RFC-0040: Install Roots & Search Paths
- RFC-0041: Mode/Topology Transition Configuration Contract
- RFC-0042: Loop Contract and UGC Bridge Semantics
- RFC-0043: Anthesis RFC Governance Model

- RFC-0044: gittuf as Git-Native Policy and Verification Substrate
- RFC-0045: Tracks as Governed Coordination Artifacts
- RFC-0046: SSDF Integration and Enforcement Model
- RFC-0047: Session Declaration Persistence and Replay Contract
- RFC-0048: Phloem Policy Evaluation and Decision Contract
- RFC-0049: Security Agent Input/Output Contract
- RFC-0050: Anthesis-Generated Commit Trailer Policy
- RFC-0051: Memory Model and Authority Rules
- RFC-0052: AIUC-1 Control Profile for Anthesis
- RFC-0053: Governed External Executor Integration
- RFC-0054: Execution Evidence and Replay Envelope
- RFC-0055: Executor Runtime Constraints
- RFC-0056: Governed Model Roles and Routing
- RFC-0057: Model Routing Resolution Mechanics
- RFC-0058: Event Forwarding and Advisory Broker Federation
- RFC-0059: Untrusted Metadata Channels in Agentic SDLC
- RFC-0060: Scope Boundary Governance and Drift Control in Anthesis
- RFC-0061: Structured Prompt Contracts
- RFC-0062: Spartan Prompt Contracts and Token-Aware Execution Profiles
- RFC-0063: Governance Signals and Signal Lifecycle
- RFC-0064: Governed Workflow Composition
- RFC-0065: Docs Routing and Bounded Context Reads
- RFC-0066: Conflict Detection and Resolution
- RFC-0067: Recursive Coordination Profiles for Governed Agent Workflows
- RFC-0068: Temporal Governance for Interaction-Derived Execution
- RFC-0099: Anthesis v1 Unfreeze & Amendment Rules

RFC-0000: Anthesis Terminology & Naming Convention

- **Status:** Living
- **Owner:** Anthesis Core
- **Created:** 2025-12-18
- **Last Updated:** 2025-12-18
- **Related RFCs:** None

This RFC canonizes Project Anthesis as the official project name and defines the shared terminology used across all existing and future RFCs.

Summary

- This RFC canonizes Project Anthesis as the official project name and defines the shared terminology used across all existing and future RFCs.

- Key sections: Canonical Project Name, Canonical Anthesis Vocabulary, RFC Naming Alignment, Backward Compatibility.

RFC-0001: Agentic SDLC Platform

- **Status:** Living
 - **Owner:** RJ
 - **Created:** 2025-01-13
 - **Last Updated:** 2026-05-07
 - **Related RFCs:** RFC-0002, RFC-0003, RFC-0004, RFC-0005, RFC-0007, RFC-0009, RFC-0017
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Summary

- Defines an agentic AI platform to support the full Software Development Lifecycle using specialized AI agents coordinated via MCP.
- Key sections: 7. Human Interaction Model, 9. Security Considerations, 2. Motivation, 3. Goals.

RFC-0002: MCP Orchestration API

- **Status:** Living
 - **Owner:** RJ
 - **Created:** 2025-01-13
 - **Last Updated:** 2026-02-10
 - **Related RFCs:** RFC-0001, RFC-0003, RFC-0004, RFC-0005, RFC-0033
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Summary

- Defines the MCP-based API for orchestrating agents and workflows across the Anthesis platform.
- Key sections: 4.5 Approval-Artifact Mapping (Normative Reference), 4.4 Proposal Taxonomy & Approval Scoping (Normative Reference), 9.1.4 Cross-Repo Query Semantics, 9.2 Cross-Org Trust Model (Artifact/Output Exchange Only).

RFC-0003: Approval Policy Engine

- **Status:** Living
- **Owner:** RJ
- **Created:** 2025-01-13
- **Last Updated:** 2026-02-11
- **Related RFCs:** RFC-0001, RFC-0002, RFC-0027, RFC-0029

Defines a declarative policy engine governing when and how agentic actions are permitted to affect the Git repository.

Summary

- Defines Calyx’s declarative policy engine for approval, risk, and governed action control.
- Requires explicit, replayable policy inputs, deterministic composition, and canonical decision outputs.
- Keeps metadata-aware validation subordinate to the existing Calyx decision vocabulary rather than inventing a parallel result model.
- Extends human review and auditability expectations when metadata interpretation safety affects policy decisions.
- Key sections: 2. Policy Model, 2.1 Policy Sources (Normative), 4. Policy Outputs, 6. Auditability.

RFC-0004: Provenance & Ingestion Model

- **Status:** Living
- **Owner:** RJ
- **Created:** 2025-01-13
- **Last Updated:** 2026-02-10
- **Related RFCs:** RFC-0001, RFC-0002, RFC-0006, RFC-0009, RFC-0023, RFC-0035

Summary

- Defines provenance tracking and data ingestion mechanisms for artifacts in Anthesis.
- Key sections: 2. Provenance Model, 2.2 Linkage Requirements, 6.1 Integrity and Append-Only Requirements, 6.2 Boundaries.

RFC-0005: Agent Schema

- **Status:** Living
 - **Owner:** RJ
 - **Created:** 2025-01-13
 - **Last Updated:** 2026-02-10
 - **Related RFCs:** RFC-0001, RFC-0002, RFC-0013, RFC-0031
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Summary

- Defines the standard schema for AI agents participating in the agentic SDLC platform.
- Key sections: 4. Execution Contract (Normative), 6. Lifecycle (Normative), 2. Agent Identity (Normative), 3. Capabilities and Declarations (Normative).

RFC-0006: Repository Event & Trigger Model

- **Status:** Living
- **Owner:** RJ
- **Created:** 2025-01-13
- **Last Updated:** 2026-02-10
- **Related RFCs:** RFC-0001, RFC-0002, RFC-0003, RFC-0004

Defines how changes to the Git repository—whether made by humans or agents—emit events that trigger agentic workflows within the MCP platform.

Summary

- Defines how changes to the Git repository—whether made by humans or agents—emit events that trigger agentic workflows within the MCP platform.
- Key sections: 4. Event Schema (Normative), 5.2 Trigger Evaluation Semantics (Normative), Future Extension: Cross-Repo Triggered Executions (Non-Normative), 3.1 Tracked Refs (Normative).

RFC-0007: Context Assembly & Prompt Composition

- **Status:** Living
- **Owner:** RJ
- **Created:** 2025-01-13
- **Last Updated:** 2026-02-10
- **Related RFCs:** RFC-0001, RFC-0002, RFC-0004, RFC-0006, RFC-0009, RFC-0030, RFC-0034, RFC-0035

Defines how the MCP server assembles contextual information and composes final prompts before invoking LLMs.

Summary

- Defines how Anthesis assembles retrieved context and composes prompts before model execution.

- Requires deterministic retrieval, explicit provenance annotation, and bounded context budgeting.
- Clarifies that display-safe metadata is not automatically prompt-safe and requires boundary-aware compartmentalization before prompt inclusion.
- Reinforces prompt security as a governed boundary, not a formatting detail.
- Key sections: 4. Retrieval & Ranking (Normative), 5. Context Budgeting (Normative), 7. Provenance Annotation (Normative), 9. Security Considerations (Normative).

RFC-0008: Model & Runtime Abstraction

- **Status:** Living
- **Owner:** RJ
- **Created:** 2025-01-13
- **Last Updated:** 2026-02-10
- **Related RFCs:** RFC-0001, RFC-0002, RFC-0005, RFC-0007, RFC-0009, RFC-0027, RFC-0035, RFC-0025

Defines an abstraction layer between agents and model runtimes, allowing MCP to select, configure, and invoke models (generation and embeddings) without coupling agents to a specific backend.

Summary

- Defines an abstraction layer between agents and model runtimes, allowing MCP to select, configure, and invoke models (generation and embeddings) without coupling agents to a specific backend.
- Key sections: 6.1 Minimal Runtime Contract (Normative), 4. Model Registry, 5. Agent-to-Model Binding, 6.2 Resolution Evidence (Normative).

RFC-0009: Agent Execution Semantics

- **Status:** Living
- **Owner:** RJ
- **Created:** 2025-12-22
- **Last Updated:** 2026-02-14
- **Related RFCs:** RFC-0001, RFC-0002, RFC-0003, RFC-0005, RFC-0007, RFC-0008

Defines how agents are executed, scheduled, retried, and terminated within the MCP platform.

Summary

- Defines agent execution, scheduling, retries, state transitions, and failure handling within Anthesis.
- Requires execution semantics to remain predictable, auditable, and aligned with the canonical Anthesis state machine.
- Prefers structured invocation over shell-like interpolation and treats unsafe boundary crossings as governed execution risks.
- Extends security guidance so boundary-specific validation or policy allowance is required before interpreter-like execution surfaces are crossed.
- Key sections: 3. Lifecycle States, 3.1 Lifecycle vs Bloom-Phase Mapping, 10.3 Execution Semantics Schema (Normative), 9. Security Considerations.

RFC-0010: CI/CD Integration Model

- **Status:** Living
- **Owner:** RJ
- **Created:** 2025-01-13
- **Last Updated:** 2026-02-10
- **Related RFCs:** RFC-0001, RFC-0002, RFC-0003, RFC-0006, RFC-0009

Defines how agentic workflows integrate with Continuous Integration and Continuous Delivery (CI/CD) systems to validate, gate, promote, and release artifacts.

Summary

- Defines how agentic workflows integrate with Continuous Integration and Continuous Delivery (CI/CD) systems to validate, gate, promote, and release artifacts.
- Key sections: 2.1 Governance Posture Alignment (Normative), 3. Integration Points, 6.1 CI Callback Events (Normative), 7.1 Multi-Environment Promotion (Normative).

RFC-0011: Observability, Telemetry & Drift

- **Status:** Living
- **Owner:** RJ
- **Created:** 2025-01-13
- **Last Updated:** 2026-02-10
- **Related RFCs:** RFC-0002, RFC-0004, RFC-0006, RFC-0007, RFC-0008, RFC-0009, RFC-0010, RFC-0023

Defines the observability framework for the agentic SDLC platform, covering logging, metrics, tracing, and drift detection across agents, models, prompts, and outcomes.

Summary

- Defines the observability framework for the agentic SDLC platform, covering logging, metrics, tracing, and drift detection across agents, models, prompts, and outcomes.
- Key sections: 3.1 Telemetry Record Envelope (Normative), 6.1 Trace Propagation (Normative), 7.1 Drift Inputs (Normative), 8. Replay & Debugging.

RFC-0012: Secrets & Credential Management

- **Status:** Living
- **Owner:** RJ
- **Created:** 2025-01-13
- **Last Updated:** 2026-02-10
- **Related RFCs:** RFC-0002, RFC-0003, RFC-0008, RFC-0009, RFC-0010, RFC-0011

Defines how secrets, credentials, and identities are managed across MCP, agents, CI/CD systems, and integrations (e.g., Git, Slack, email).

Summary

- Defines how secrets, credentials, and identities are managed across MCP, agents, CI/CD systems, and integrations (e.g., Git, Slack, email).
- Key sections: 6.1 Per-Execution Secret Access (Normative), 8.1 Secret Audit Envelope (Normative), 3. Identity Model, 4.1 Secret Descriptor (Normative).

RFC-0013: Multi-Agent Coordination & Chaining

- **Status:** Living
- **Owner:** RJ
- **Created:** 2025-01-13
- **Last Updated:** 2026-05-21
- **Related RFCs:** RFC-0002, RFC-0005, RFC-0006, RFC-0007, RFC-0009, RFC-0010, RFC-0011, RFC-0064, RFC-0067

Defines how multiple agents collaborate within a single workflow, including sequencing, fan-out/fan-in patterns, dependency management, and deadlock prevention.

Summary

- Defines how multiple agents collaborate within a single workflow, including sequencing, fan-out/fan-in patterns, dependency management, and deadlock prevention.
- Key sections: 4.2 Workflow Graph Artifact (Normative), 5.1 Execution Linkage (Normative), 6. Approval Boundaries, 6.1 Workflow-Level Bounds (Normative).

RFC-0014: Risk Classification & Bloom Classes

- **Status:** Living
- **Owner:** RJ
- **Created:** 2025-01-13
- **Last Updated:** 2026-02-10
- **Related RFCs:** RFC-0003, RFC-0006, RFC-0007, RFC-0009, RFC-0010, RFC-0011, RFC-0013, RFC-0020

Defines a unified risk model used across the agentic SDLC platform to govern approvals, execution semantics, context limits, CI/CD gates, and human oversight.

Summary

- Defines a unified risk model used across the agentic SDLC platform to govern approvals, execution semantics, context limits, CI/CD gates, and human oversight.
- Key sections: 5.2 Authority and Timing (Normative), 5.3 Override Semantics (Normative), 5.1 Risk Computation Inputs (Normative), 9. Security Considerations.

RFC-0015: Human Interfaces & Control Surfaces

- **Status:** Living
- **Owner:** RJ
- **Created:** 2026-01-18
- **Last Updated:** 2026-02-10
- **Related RFCs:** RFC-0002, RFC-0003, RFC-0009, RFC-0014, RFC-0021

Defines required human control surfaces for Anthesis, including approval interactions, visibility, and safe intervention boundaries.

Summary

- Defines required human control surfaces for Anthesis, including approval interactions, visibility, and safe intervention boundaries.
- Key sections: **Approval & Pruning Model**, **Repo Scoping (Normative)**, **Approval Actions**, **Visualization & Canonical Projections (Normative, v1)**.

RFC-0016: Dormancy & Disaster Recovery

- **Status:** Living
- **Owner:** Anthesis Core Team
- **Created:** 2025-12-18
- **Last Updated:** 2026-02-10
- **Related RFCs:** RFC-0003, RFC-0004, RFC-0009, RFC-0010, RFC-0011, RFC-0012, RFC-0014, RFC-0015, RFC-0021, RFC-0023

This RFC defines how Project Anthesis behaves under failure, interruption, corruption, or emergency conditions. It introduces Dormancy as a first-class system state and establishes standards for backup, recovery, replay, and graceful degradation.

Summary

- This RFC defines how Project Anthesis behaves under failure, interruption, corruption, or emergency conditions. It introduces Dormancy as a first-class system state and establishes standards for backup, recovery, replay, and graceful degradation.
- Key sections: **Boundaries (Normative)**, **In-Flight Execution Handling (Normative, v1)**, **Replay Retention Window (Normative, v1)**, **Scope (Normative)**.

RFC-0017: Governance, Ownership & RFC Lifecycle

- **Status:** Living
- **Owner:** Anthesis Core
- **Created:** 2025-12-18
- **Last Updated:** 2026-05-07
- **Related RFCs:** RFC-0003, RFC-0004, RFC-0014, RFC-0015, RFC-0021, RFC-0023, RFC-0099, CHARTER.md

This RFC defines how Project Anthesis governs itself. It establishes ownership models, decision authority, and the lifecycle for RFCs, agents, policies, and models.

Summary

- This RFC defines how Project Anthesis governs itself. It establishes ownership models, decision authority, and the lifecycle for RFCs, agents, policies, and models.
- Key sections: Ownership Model, Ownership Recording (Normative), Roles and Authority, RFC Lifecycle.

RFC-0018: Anthesis Tooling & Project Bootstrap

- **Status:** Living
- **Owner:** Anthesis Core
- **Created:** 2025-12-18
- **Last Updated:** 2026-02-12
- **Related RFCs:** RFC-0000, RFC-0001, RFC-0002, RFC-0003, RFC-0006, RFC-0010, RFC-0014, RFC-0015, RFC-0016, RFC-0017, RFC-0021, RFC-0023

This RFC defines the tooling layer that applies Anthesis to new and existing projects. It introduces a CLI-driven bootstrap and adoption model that standardizes governance while preserving project autonomy.

Summary

- This RFC defines the tooling layer that applies Anthesis to new and existing projects. It introduces a CLI-driven bootstrap and adoption model that standardizes governance while preserving project autonomy.
- Key sections: Runtime and State (Normative, v1), Plugin System (Normative, v1), Version Pinning (Normative, v1), Integration with n8n and MCP.

RFC-0019: Anthesis Core Distribution & Update Strategy

- **Status:** Living
- **Owner:** Anthesis Core
- **Created:** 2025-12-18
- **Last Updated:** 2026-02-10
- **Related RFCs:** RFC-0000, RFC-0003, RFC-0014, RFC-0017, RFC-0018, RFC-0021, RFC-0023

This RFC defines how Anthesis Core (shared agents, policies, profiles, and templates) is distributed, versioned, and updated across multiple projects while preserving local autonomy and auditability.

Summary

- This RFC defines how Anthesis Core (shared agents, policies, profiles, and templates) is distributed, versioned, and updated across multiple projects while preserving local autonomy and auditability.
- Key sections: `Boundaries (Normative)`, `Provenance & Audit`, `Provenance Recording Location (Normative, v1)`, `Distribution Models`.

RFC-0020: Anthesis State Machine (Formal Specification)

- **Status:** Living
- **Owner:** Anthesis Core
- **Created:** 2025-12-18
- **Last Updated:** 2026-02-10
- **Related RFCs:** RFC-0000, RFC-0003, RFC-0007, RFC-0009, RFC-0014, RFC-0016

This RFC formally specifies the Anthesis State Machine, defining the canonical workflow phases, transitions, invariants, and failure behavior for agentic workflows.

Summary

- This RFC formally specifies the Anthesis State Machine, defining the canonical workflow phases, transitions, invariants, and failure behavior for agentic workflows.
- Key sections: `Boundaries (Normative)`, `State Transitions`, `Replay Semantics`, `Canonical Workflow Phases`.

RFC-0021: Anthesis CLI UX & Command Semantics

- **Status:** Living
- **Owner:** Anthesis Core
- **Created:** 2025-01-05
- **Last Updated:** 2026-02-19
- **Related RFCs:** RFC-0000, RFC-0008, RFC-0009, RFC-0015, RFC-0018, RFC-0020, RFC-0025, RFC-0027, RFC-0035

This RFC specifies the command-line interface (CLI) for Project Anthesis. The CLI is a first-class human control surface providing deterministic, scriptable access to Anthesis workflows while preserving governance, approvals, and safety invariants.

Summary

- This RFC specifies the command-line interface (CLI) for Project Anthesis. The CLI is a first-class human control surface providing deterministic, scriptable access to Anthesis workflows while preserving governance, approvals, and safety invariants.
- Key sections: Workflow Phase vs Execution State (Normative), 3) Approval Flow (If Required), 5) Audit & Execution Reads (Optional CLI Surface), anthesis model benchmark.

RFC-0022: External Integrations & MCP Federation

- **Status:** Living
- **Owner:** Anthesis Core
- **Created:** 2025-12-18
- **Last Updated:** 2026-02-10
- **Related RFCs:** RFC-0000, RFC-0002, RFC-0006, RFC-0015, RFC-0017, RFC-0018, RFC-0020, RFC-0021, RFC-0023

This RFC defines how Project Anthesis integrates with external systems and how multiple MCP servers may federate securely.

Summary

- This RFC defines how Project Anthesis integrates with external systems and how multiple MCP servers may federate securely.
- Key sections: Boundaries (Normative), Integration Principles, Integration Classes, Inbound Integrations.

RFC-0023: Audit, Compliance & Evidence Export

- **Status:** Living
- **Owner:** Anthesis Core
- **Created:** 2025-12-18
- **Last Updated:** 2026-02-10
- **Related RFCs:** RFC-0000, RFC-0004, RFC-0011, RFC-0014, RFC-0016, RFC-0020, RFC-0022

This RFC defines how Project Anthesis produces, preserves, and exports audit evidence suitable for internal review, incident forensics, and external compliance frameworks (e.g., SOC 2, ISO 27001).

Summary

- This RFC defines how Project Anthesis produces, preserves, and exports audit evidence suitable for internal review, incident forensics, and external compliance frameworks (e.g., SOC 2, ISO 27001).
- Key sections: `Boundaries` (Normative), `External Auditor Access` (Normative, v1), `Audit Views`, `Evidence Bundle Manifest` (Normative, v1).

RFC-0024: Anthesis UI & Visualization Standards

- **Status:** Living
- **Owner:** Anthesis Core
- **Created:** 2025-12-18
- **Last Updated:** 2026-02-10
- **Related RFCs:** RFC-0000, RFC-0009, RFC-0011, RFC-0013, RFC-0015, RFC-0020, RFC-0021, RFC-0023

This RFC defines user interface (UI) and visualization standards for Project Anthesis. It specifies how system state, agent behavior, policies, risk, and evidence are rendered for humans in a way that is accurate, explainable, and safe.

Summary

- This RFC defines user interface (UI) and visualization standards for Project Anthesis. It specifies how system state, agent behavior, policies, risk, and evidence are rendered for humans in a way that is accurate, explainable, and safe.
- Key sections: `Workflow Phase vs Execution State` (Normative), `2. Execution Timeline View`, `4. Approval Queue`, `5. Audit & Evidence Explorer`.

RFC-0025: Model Evaluation, Benchmarking & Drift Budgets

- **Status:** Living
- **Owner:** Anthesis Core
- **Created:** 2025-12-18
- **Last Updated:** 2026-05-20
- **Related RFCs:** RFC-0000, RFC-0007, RFC-0008, RFC-0011, RFC-0014, RFC-0020, RFC-0021, RFC-0023, RFC-0052, RFC-0054, RFC-0063, RFC-0067

This RFC defines how Project Anthesis evaluates language models, benchmarks their performance, and enforces drift budgets to ensure reliability, safety, and cost control over time.

Summary

- This RFC defines how Project Anthesis evaluates language models, benchmarks their performance, and enforces drift budgets to ensure reliability, safety, and cost control over time.
- Key sections: `Boundaries (Normative)`, `Evaluation Evidence Requirements (Normative, v1)`, `Drift Evaluation Rules (Normative, v1)`, `Drift Breach Actions (Normative, v1)`.

RFC-0026: Anthesis Plugin & Extension Model

- **Status:** Living
- **Owner:** Anthesis Core
- **Created:** 2025-12-18
- **Last Updated:** 2026-02-10
- **Related RFCs:** RFC-0000, RFC-0002, RFC-0003, RFC-0014, RFC-0015, RFC-0017, RFC-0018, RFC-0021, RFC-0022

This RFC defines a plugin and extension model for Project Anthesis, enabling third-party and internal extensions while preserving governance, safety, and auditability.

Summary

- This RFC defines a plugin and extension model for Project Anthesis, enabling third-party and internal extensions while preserving governance, safety, and auditability.
- Key sections: `Boundaries (Normative)`, `Manifest Minimum Contract (Normative, v1)`, `Security Model`, `Sandboxing Requirements (Normative, v1)`.

RFC-0027: Configuration Model

- **Status:** Living
- **Owner:** RJ
- **Created:** 2025-12-28
- **Last Updated:** 2026-02-14
- **Related RFCs:** RFC-0002, RFC-0003, RFC-0012, RFC-0018, RFC-0021, RFC-0040, RFC-0041

Defines how Anthesis services load configuration across multiple processes and environments, with explicit precedence and security guarantees.

Summary

- Defines Anthesis configuration sources, precedence, and security constraints across services and environments.
- Requires deterministic profile resolution, explicit process boundaries, and startup validation for governed runtime behavior.
- Clarifies that metadata interpretation-safety behavior must be configured explicitly, validated at load time, and fail closed when invalid.
- Adds auditability expectations for resolved configuration provenance.
- Key sections: 5. Configuration Sources, 5.1 CLI Session Profile Resolution (Normative, v1), 9. Validation and Auditability, 12. Transitional Runtime Mode/Topology Contract (Normative, v1).

RFC-0028: Repo Registry & Identity Resolution

- **Status:** Living
 - **Owner:** Anthesis Core
 - **Created:** 2026-02-11
 - **Last Updated:** 2026-02-19
 - **Related RFCs:** RFC-0002, RFC-0006, RFC-0010, RFC-0029, RFC-0035
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Defines how Anthesis identifies repositories (`repo_id`) and resolves multiple aliases (`https://`, `ssh://`, `file://`, local paths) to one canonical identity.

Summary

- Defines how Anthesis identifies repositories (`repo_id`) and resolves multiple aliases (`https://`, `ssh://`, `file://`, local paths) to one canonical identity.
- Key sections: 4) Alias Lifecycle Controls (Normative, v1), 5) Evidence Requirements for Alias Changes (Normative, v1), 6) Reachability/Deletion Semantics (Normative, v1), v1 Decisions (Normative).

RFC-0029: Policy Composition & Override Resolution

- **Status:** Living
- **Owner:** Anthesis Core
- **Created:** 2026-02-11
- **Last Updated:** 2026-02-19
- **Related RFCs:** RFC-0002, RFC-0003, RFC-0027, RFC-0028, RFC-0033, RFC-0035

Defines how Anthesis composes policy from organization baseline controls and repo-level overrides, including fail-closed conflict handling.

Summary

- Defines how Anthesis composes policy from organization baseline controls and repo-level overrides, including fail-closed conflict handling.
- Key sections: 7) Effective Policy Export Contract (Normative, v1), 1) Composition Model (Normative, v1), 2) Configuration Source Precedence (Normative, v1), v1 Decisions (Normative).

RFC-0030: Approval-Artifact Mapping

- **Status:** Living
- **Owner:** RJ
- **Created:** 2025-01-13
- **Last Updated:** 2026-02-10
- **Related RFCs:** RFC-0002, RFC-0003, RFC-0004, RFC-0007, RFC-0023, RFC-0033

Defines how approval decisions map to specific artifacts (path + commit) so that context blocks can report deterministic, auditable approval_status metadata.

Summary

- Maps approvals to exact artifacts using path plus commit identity.
- Prevents vague or overbroad approval claims from leaking across changes.
- Supports deterministic approval-status reporting in context assembly and audit surfaces.
- Key sections: 8. Provenance & Audit Alignment (RFC-0004 / RFC-0023), 5. Data Model (Logical), 5.2 Approval-Artifact Link, 6. Mapping Rules.

RFC-0031: Agent Classes & Trust Model

- **Status:** Living
- **Owner:** Anthesis Core
- **Created:** 2026-01-17
- **Last Updated:** 2026-02-10
- **Related RFCs:** RFC-0003, RFC-0005, RFC-0013, RFC-0014, RFC-0023

Defines canonical Anthesis agent classes and trust boundaries, separating authority from execution with explicit non-escalation constraints.

Summary

- Defines canonical Anthesis agent classes and trust boundaries, separating authority from execution with explicit non-escalation constraints.
- This RFC gives Anthesis an authoritative reference for canonical Anthesis agent classes and trust boundaries, separating authority from execution with explicit non-escalation constraints.
- Without that reference, the same concern is likely to drift across services, tools, or workflows.
- Key sections: **Action Authority Matrix (Normative, v1)**, **Authority Flow (Hard Rule)**, **Canonical Class IDs (Normative, v1)**, **Calyx - Authority & Accountability**.

RFC-0032: External Tool Integration & Admission

- **Status:** Living
- **Owner:** Anthesis Core
- **Created:** 2026-01-17
- **Last Updated:** 2026-05-21
- **Related RFCs:** RFC-0003, RFC-0014, RFC-0022, RFC-0023, RFC-0026, RFC-0031, RFC-0049, RFC-0054

Defines admission controls and governance requirements for external tools and agent integrations, including policy, auditability, and trust-boundary constraints.

Summary

- Defines how external tools and agent integrations may be admitted into Anthesis.
- Requires explicit policy, scope, provenance, and auditability constraints.
- Treats integrations as trust-boundary crossings rather than neutral helpers.
- Requires execution-capable tools to satisfy boundary-safe invocation expectations in addition to admission and provenance controls.
- Anthesis is agentic and tool-capable, so unmanaged integrations can silently expand authority and risk.
- This RFC keeps external tool use governed, attributable, and revocable.
- Key sections: **Admission Contract (Normative, v1)**, **Tool Participation Model**, **Admission Lifecycle (Normative, v1)**, **Provenance Requirements**.

RFC-0033: Proposal Taxonomy & Approval Scoping Strategy

- **Status:** Living

- **Owner:** Anthesis Core
- **Created:** 2026-02-05
- **Last Updated:** 2026-02-19
- **Related RFCs:** RFC-0002, RFC-0003, RFC-0007, RFC-0020, RFC-0023, RFC-0030

This RFC defines a minimal, normative proposal taxonomy and the rules that govern approval scoping in Anthesis.

Summary

- Defines the proposal kinds Anthesis recognizes.
- Limits which proposal types may confer artifact-level approval scope.
- Prevents execution or operational approvals from being misread as artifact authorization.
- Approval scope is a governance boundary, not a convenience label.
- This RFC prevents Anthesis from inferring artifact approval from the wrong proposal type and keeps approval semantics reviewable.
- Key sections: 0. Boundaries (Normative), 6. Approval Scoping Rules (Normative), 6.2 Execution Gate Semantics, 5. Proposal Taxonomy (Normative).

RFC-0034: Sessions & Prompt Context Governance

- **Status:** Living
- **Owner:** Anthesis Core
- **Created:** 2026-01-26
- **Last Updated:** 2026-02-11
- **Related RFCs:** RFC-0000, RFC-0004, RFC-0007, RFC-0009, RFC-0021, RFC-0023, RFC-0027, RFC-0030

This RFC defines sessions and the governance rules for prompt context in Anthesis. It introduces a machine-readable session declaration, the Core Invariant vs Session Overlay model, and provenance expectations required to reconstruct the Effective Prompt after execution.

Summary

- Defines sessions as governed prompt-context declarations.
- Separates always-on core invariants from narrower session overlays.
- Requires machine-readable session declarations and provenance sufficient to reconstruct effective prompt context.
- Key sections: 4.1 Boundaries (Normative), 7.3.1 Minimum Policy Snapshot (Normative, v1), 8. Audit & Provenance, 9. CLI

Integration (Normative Reference).

RFC-0035: Operating Modes (Offline / Local / Remote)

- **Status:** Living
- **Owner:** RJ
- **Created:** 2026-01-29
- **Last Updated:** 2026-02-14
- **Related RFCs:** None

This RFC defines Anthesis operating modes—offline, local, and remote—and the constraints, service dependencies, and governance expectations that apply to each. The goal is to make runtime posture explicit, predictable, and auditable while preserving RFC-level invariants (policy enforcement, auditability, and safe defaults).

Summary

- Defines offline, local, and remote operating modes.
- Makes runtime posture explicit in terms of available services and governance constraints.
- Preserves Anthesis invariants even when parts of the system are unavailable.
- Key sections: `Transitions (Normative)`, `Transition Safety and Recoverability (Normative)`, `Governance Requirements by Mode (Normative)`, `Runtime Axes (Normative, v1)`.

RFC-0036: QART Engineering

- **Status:** Living
- **Owner:** Governance Layer
- **Created:** 2026-02-21
- **Last Updated:** 2026-02-23
- **Related RFCs:** RFC-0037, RFC-0038, RFC-0099

This RFC defines QART Engineering as the required pre-decision governance discipline for significant architectural intent.

Summary

- This RFC defines QART Engineering as the required pre-decision governance discipline for significant architectural intent.
- Key sections: `5. The Four Phases (Normative Specification)`, `8. Escalation Model`, `9. Compliance Requirements`, `12. AI-Agent Participation Policy`.

RFC-0037: Drift Loop Engineering

- **Status:** Living
 - **Owner:** Governance Layer
 - **Created:** 2026-02-21
 - **Last Updated:** 2026-05-21
 - **Related RFCs:** RFC-0036, RFC-0038, RFC-0051, RFC-0099
-

This RFC defines Drift Loop Engineering as the governed control loop for detecting, classifying, reconciling, and auditing drift between canonical intent and implementation reality.

Summary

- This RFC defines Drift Loop Engineering as the governed control loop for detecting, classifying, reconciling, and auditing drift between canonical intent and implementation reality.
- Key sections: 5. Normative Requirements, 14. Drift Lifecycle State Machine, 4. System Model, 4.2 Loop Lifecycle.

RFC-0038: Unified Governance Control Doctrine

- **Status:** Living
 - **Owner:** Anthesis Core
 - **Created:** 2026-02-21
 - **Last Updated:** 2026-05-07
 - **Related RFCs:** RFC-0036, RFC-0037, RFC-0042, RFC-0099
-

This RFC defines Unified Governance Control as the meta-doctrine that binds QART pre-decision governance with Drift Loop post-decision governance.

Summary

- Binds QART and Drift Loop into one governance doctrine.
- Connects pre-decision review with post-decision drift handling.
- Defines governance as a continuous control system rather than isolated checks.
- Key sections: 5. Governance Architecture, 7. Control Properties, 9. AI-Agent Governance, 10. Failure Modes.

RFC-0039: MCP Transport Profiles (`http`, `stdio`, `sse`)

- **Status:** Living
- **Owner:** TBD
- **Created:** 2026-02-22

- **Last Updated:** 2026-02-22
- **Related RFCs:** RFC-0002, RFC-0003, RFC-0017, RFC-0021, RFC-0027, RFC-0035, RFC-0099

This RFC defines governed MCP transport profile support in Phloem across `http`, `stdio`, and `sse` with explicit mode and security constraints.

Summary

- This RFC defines governed MCP transport profile support in Phloem across `http`, `stdio`, and `sse` with explicit mode and security constraints.
- Key sections: 6. Security Requirements (Proposed Normative), 8. Configuration Contract (Proposed Normative), 7.2 Startup Validation Matrix (Normative, v1), 10. Audit and Evidence Requirements.

RFC-0040: Install Roots & Search Paths

- **Status:** Living
- **Owner:** Anthesis Core
- **Created:** 2026-03-10
- **Last Updated:** 2026-03-10
- **Related RFCs:** RFC-0017, RFC-0018, RFC-0021, RFC-0027, RFC-0028, RFC-0034

This RFC defines governed install roots and deterministic search path resolution for Anthesis prompt and session assets.

Summary

- This RFC defines governed install roots and deterministic search path resolution for Anthesis prompt and session assets.
- Key sections: 11. Security Model (Normative), 4.3 Auditability and Provenance Requirement, 10. Contract (Normative), 10.6 Provenance and Output Requirements.

RFC-0041: Mode/Topology Transition Configuration Contract

- **Status:** Living
- **Owner:** Anthesis Core
- **Created:** 2026-03-10
- **Last Updated:** 2026-03-10
- **Related RFCs:** RFC-0017, RFC-0021, RFC-0027, RFC-0035, RFC-0039

This RFC defines the configuration contract for operating mode and execution topology transitions, extracting the transition-specific posture rules from RFC-0027 into a dedicated control surface.

Summary

- Defines the config contract for mode and topology transitions.
- Separates transition posture rules from general configuration concerns.
- Requires reproducible, auditable, fail-closed transition behavior.
- Key sections: 4. Normative Contract (Extracted from RFC-0027 §12), 4.3 Runtime Validation Requirements, 4.4 Worker Topology Queue Configuration, 4.6 Migration-Phase Contract.

RFC-0042: Loop Contract and UGC Bridge Semantics

- **Status:** Living
- **Owner:** Anthesis Core
- **Created:** 2026-03-17
- **Last Updated:** 2026-03-17
- **Related RFCs:** RFC-0009, RFC-0013, RFC-0017, RFC-0020, RFC-0021, RFC-0022, RFC-0036, RFC-0037, RFC-0038

This RFC defines a loop contract for Inflorescence-managed loop types and a governed bridge model for relating loop instances under Unified Governance Control, without changing Anthesis workflow-phase law or execution lifecycle semantics.

Summary

- Defines governed loop and bridge semantics for Inflorescence-managed lifecycle artifacts.
- Relates loop instances without changing Anthesis execution law.
- Keeps loop coordination subordinate to existing workflow and governance primitives.
- Key sections: 16. Authority Boundaries, 6. Loop Contract, 7. Bridge Contract, 8. Loop Types And Stage Semantics.

RFC-0043: Anthesis RFC Governance Model

- **Status:** Review
- **Owner:** RJ
- **Created:** 2026-03-22
- **Last Updated:** 2026-03-22
- **Related RFCs:** RFC-0017, RFC-0099

This RFC defines the lifecycle, mutability model, and governance rules for RFCs within Anthesis, treating RFCs as governed, versioned, machine-readable system artifacts rather than static documents.

Summary

- Defines the lifecycle and mutability model for Anthesis RFCs.
- Treats RFCs as governed system artifacts rather than static prose.
- Sets expectations for amendment, transition, and enforcement behavior.
- Key sections: Lifecycle Model, Core Mental Model, States, Mutability Model.

RFC-0044: gittuf as Git-Native Policy and Verification Substrate

- **Status:** Living
- **Owner:** Anthesis Core
- **Created:** 2026-03-26
- **Last Updated:** 2026-03-26
- **Related RFCs:** RFC-0003, RFC-0004, RFC-0010, RFC-0016, RFC-0017, RFC-0021, RFC-0023, RFC-0028, RFC-0030, RFC-0032, RFC-0033, RFC-0035

This RFC defines how Anthesis integrates gittuf as the Git-native policy enforcement and verification substrate for protected refs, protected paths, and approval-backed repository changes without replacing Anthesis governance.

Summary

- Uses gittuf as the repository-native verification and policy layer.
- Keeps Anthesis governance, approvals, and evidence semantics authoritative.
- Protects refs, paths, and release workflows with portable verification.
- Bridges Anthesis approvals into Git-native attestations without making gittuf the governance system.
- Key sections: 6.1 Trust Model, 6.2 Policy Model, anthesis trust attest-approval, 6.3 Approval Bridging.

RFC-0045: Tracks as Governed Coordination Artifacts

- **Status:** Living
- **Owner:** RJ
- **Created:** 2026-03-19
- **Last Updated:** 2026-03-26

- **Related RFCs:** RFC-0002, RFC-0009, RFC-0020, RFC-0023, RFC-0033, RFC-0034, RFC-0038

This RFC defines tracks as governed, repo-scoped coordination artifacts that group references to authoritative Anthesis artifacts for bounded streams of work, without introducing new execution authority, approval scope, or runtime state.

Summary

- Defines tracks as reference-oriented coordination artifacts only.
- Lets Anthesis group tasks, sessions, approvals, executions, and evidence for bounded workstreams.
- Prohibits tracks from becoming execution containers, approval grants, or runtime state objects.
- Key sections: 4. Authority Boundaries, 9. Lifecycle and Removal Semantics, 14. Audit Model, 3. Normative Basis.

RFC-0046: SSDF Integration and Enforcement Model

- **Status:** Living
- **Owner:** Anthesis Core
- **Created:** 2026-03-22
- **Last Updated:** 2026-03-26
- **Related RFCs:** RFC-0002, RFC-0003, RFC-0009, RFC-0010, RFC-0020, RFC-0023, RFC-0032, RFC-0038, RFC-0044, RFC-0045

This RFC defines how NIST SSDF is integrated into Anthesis as a governed security baseline and enforcement model across policy, evidence, scanning, artifact validation, and vulnerability response, without redefining runtime authority.

Summary

- Maps SSDF into Anthesis as governed security requirements.
- Enforces security through policy, validation, evidence, and response loops.
- Keeps SSDF subordinate to Anthesis governance rather than creating a parallel compliance system.
- Uses existing Anthesis primitives instead of inventing new execution states or pipelines.
- Key sections: 9. Audit Model, 5. Normative Basis, 6. SSDF Mapping, 7. Enforcement Model.

RFC-0047: Session Declaration Persistence and Replay Contract

- **Status:** Living
 - **Owner:** Anthesis Core
 - **Created:** 2026-03-26
 - **Last Updated:** 2026-03-26
 - **Related RFCs:** RFC-0020, RFC-0023, RFC-0034
-

Summary

- Stores Anthesis session declarations as immutable Phloem records.
- Requires append-only linkage from sessions to executions.
- Makes server-side normalized hashing authoritative.
- Supports replay as attributable context-lineage reconstruction.
- Leaves evidence packaging and retention under RFC-0023.
- Key sections: 7. Replay Semantics, 4. Session Record Model, 6. Session-to-Execution Association, 1. Purpose.

RFC-0048: Phloem Policy Evaluation and Decision Contract

- **Status:** Living
 - **Owner:** Anthesis Core
 - **Created:** 2026-03-26
 - **Last Updated:** 2026-03-26
 - **Related RFCs:** RFC-0003, RFC-0023, RFC-0030, RFC-0032, RFC-0033, RFC-0044, RFC-0045, RFC-0046
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Summary

- Defines Phloem's normalized evaluation context and policy result vocabulary.
- Requires strictest-wins aggregation and machine-readable explanations.
- Defines the PDR as the authoritative Phloem decision record.
- Allows PDR-derived attestations without turning them into approval artifacts.
- Extends evaluation and replay context to carry metadata-validation results when policy decisions depend on interpretation safety.
- Keeps approval, evidence, and Git-native verification semantics in their existing canonical RFCs.
- Key sections: 4. Normalized Evaluation Context, 5. Policy Results and Aggregation, 6. Evaluator Explanation Contract, 7.3 Decision Model.

RFC-0049: Security Agent Input/Output Contract

- **Status:** Living
 - **Owner:** Anthesis Core
 - **Created:** 2026-03-26
 - **Last Updated:** 2026-05-21
 - **Related RFCs:** RFC-0032, RFC-0045, RFC-0046, RFC-0048, RFC-0054
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Summary

- Defines the bounded producer contract for Anthesis security agents.
- Standardizes review inputs, verdicts, findings, and evidence summaries.
- Keeps agent output advisory to Phloem rather than gate-authoritative.
- Prevents agent findings from replacing human approval or deterministic controls.
- Key sections: 6. Output Contract, 1. Purpose, 2. Scope, In Scope.

RFC-0050: Anthesis-Generated Commit Trailer Policy

- **Status:** Living
 - **Owner:** Anthesis Core
 - **Created:** 2026-03-26
 - **Last Updated:** 2026-03-26
 - **Related RFCs:** RFC-0004, RFC-0006, RFC-0011, RFC-0023, RFC-0030, RFC-0032, RFC-0033, RFC-0034, RFC-0044
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Summary

- Defines an optional advisory trailer for Anthesis-generated commits.
- Keeps trailer metadata subordinate to governed evidence and RFC-0044 verification.
- Prohibits inferring approval or authorization from trailer presence alone.
- Defines corroboration, precedence, and history-rewrite handling rules.
- Key sections: 5. Consumer Semantics, 6. Security and Compliance, 1. Purpose, 2. Scope.

RFC-0051: Memory Model and Authority Rules

- **Status:** Living
 - **Owner:** Anthesis Core
 - **Created:** 2026-03-26
 - **Last Updated:** 2026-05-21
 - **Related RFCs:** RFC-0004, RFC-0007, RFC-0023, RFC-0054, RFC-0099
-

Summary

- Defines Anthesis memory classes, authority order, read/write rules, conflict handling, and minimum provenance and security constraints for durable memory.
- Prevents chat history or retrieved context from silently becoming normative system state.
- Requires provenance, conflict handling, and explicit mutation paths for durable memory.
- Treats memory as governable context that must remain auditable and externally verifiable.
- Key sections: 5. Memory Classes, 6. Authority and Precedence, 7. Read Policy, 8. Write Policy.

RFC-0052: AIUC-1 Control Profile for Anthesis

- **Status:** Living
 - **Owner:** Anthesis Core
 - **Created:** 2026-04-10
 - **Last Updated:** 2026-05-20
 - **Related RFCs:** RFC-0002, RFC-0003, RFC-0009, RFC-0017, RFC-0023, RFC-0025, RFC-0027, RFC-0032, RFC-0036, RFC-0038, RFC-0042, RFC-0046, RFC-0049
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Summary

- Defines how Anthesis may admit and evaluate AIUC-1 profile releases as subordinate governed control artifacts.
- Keeps AIUC-1 as a profile layer, not a parallel governance or execution authority.
- Requires AIUC-1-aware policy, execution, and evidence handling to remain mediated through existing Anthesis RFC surfaces.
- Fails closed when profile metadata, applicability, or release constraints are incomplete.
- Key sections: 3. Governance Position, 4. Normative Basis, 8. Control Profile Model, 11. Evidence Model.

RFC-0053: Governed External Executor Integration

- **Status:** Living
 - **Owner:** Unknown
 - **Created:** 2026-04-16
 - **Last Updated:** Unknown
 - **Related RFCs:** None
-

Defines how Anthesis may admit an external execution runtime without creating a second control plane or weakening Anthesis governance.

Summary

- Treats external executors as admitted tools under Anthesis control, not as governance authorities.
- Requires Anthesis to own intent binding, authorization, approval semantics, and evidence interpretation.
- Keeps executor-native capability labels subordinate to Anthesis-owned execution law.
- Requires explicit admission, scoped capability limits, and attributable evidence for every run.
- Distinguishes adapter responsibilities from executor implementation details.
- Key sections: Scope, Stable Principles, Definitions, Executor Invocation Spec, Adapter Responsibilities, Decision Record, Acceptance Outcome.

RFC-0054: Execution Evidence and Replay Envelope

- **Status:** Living
- **Owner:** RJ
- **Created:** 2026-04-16
- **Last Updated:** 2026-05-21
- **Related RFCs:** RFC-0002, RFC-0007, RFC-0009, RFC-0011, RFC-0023, RFC-0025, RFC-0030, RFC-0047, RFC-0053, RFC-0067, RFC-0068

Defines a canonical evidence and replay envelope for Anthesis executions, including executions performed by admitted external executors.

Summary

- Defines a stable evidence envelope for governed executions and later replay or re-evaluation.
- Separates authoritative Anthesis records from non-authoritative executor transcripts.
- Requires Anthesis to capture what was asked, what authority was granted, what happened, and what outputs were produced.
- Distinguishes execution-level replay from session declaration replay law.
- Keeps the envelope complementary to existing audit, provenance, and session persistence RFCs.
- Key sections: Relationship to RFC-0047, Envelope Model, Normative Requirements, Replayability Classes, Relationship to Audit and Provenance, Relationship to External Executors.

RFC-0055: Executor Runtime Constraints

- **Status:** Living
- **Owner:** Unknown
- **Created:** 2026-04-16
- **Last Updated:** Unknown
- **Related RFCs:** None

Defines how admitted external executors must conform to Anthesis runtime posture, rather than introducing a second sandbox taxonomy.

Summary

- Reuses Anthesis runtime axes as the top-level authority for executor posture.
- Requires effective capability scope to be policy-derived, not executor-derived.
- Keeps executor-native mode labels descriptive and subordinate to Anthesis runtime law.
- Narrows interactive and chat-adjacent capability surfaces unless explicitly elevated.
- Maps executor behavior into Anthesis operating modes and capability classes.
- Key sections: Normative Requirements, Operating-Mode Expectations, Capability Classes, Interactive Surface Constraints, Mapping Guidance, Acceptance Outcome.

RFC-0056: Governed Model Roles and Routing

- **Status:** Living
- **Owner:** Unknown
- **Created:** 2026-04-18
- **Last Updated:** Unknown
- **Related RFCs:** None

Defines the governing contract for how Anthesis model roles, aliases, and routing behave across loops, control surfaces, and execution stages.

Summary

- Defines stable governance roles for Anthesis model participation across planning, execution, review, arbitration, and evidence loops.
- Treats routing policy as governed, file-backed, and auditable rather than implicit chat-state behavior.

- Keeps the AI client as an operator surface, not the source of truth for governance, routing, or approval state.
- Requires alias bindings to be explicit and reconstructable from governed configuration.
- Records the open routing decisions in a formal decision record and delegates routing mechanics to RFC-0057.
- Key sections: 5. Routing Governance, 6. Model roles in Anthesis loops, 9. Role of the AI client, 14. Artifact expectations per loop, 16. Failure handling, 18. Decision Record.

RFC-0057: Model Routing Resolution Mechanics

- **Status:** Living
- **Owner:** Unknown
- **Created:** 2026-04-18
- **Last Updated:** Unknown
- **Related RFCs:** None

Defines how Anthesis resolves governed model requests into concrete runtime bindings, including fallback behavior, degraded routing, and audit evidence.

Summary

- Defines deterministic resolution order for governed model requests.
- Allows capacity-aware remapping only when policy permits it and the degraded choice remains auditable.
- Requires each resolved binding to identify the backend or provider type, the deployment or model identifier, and the capability match rationale.
- Records replayable audit evidence for each resolution so the decision can be reconstructed later.
- Keeps routing mechanics separate from the governance contract in RFC-0056.
- Key sections: 4. Resolution Order, 5. Capacity-Aware Remapping, 6. Backend and Provider Binding, 7. Audit Record, 8. Failure Handling, 10. Relationship to RFC-0056.

RFC-0058: Event Forwarding and Advisory Broker Federation

- **Status:** Living
- **Owner:** Unknown
- **Created:** 2026-04-18
- **Last Updated:** Unknown
- **Related RFCs:** None

Defines how Anthesis may forward selected execution and task events across broker boundaries without creating a second source of truth.

Summary

- Treats forwarded events as advisory notifications rather than authoritative state.
- Requires explicit allowlists for forwarding scope and trust-boundary preservation.
- Requires a recoverable forwarded-event envelope with source identity, event identity, route identity, and audit attribution.
- Requires a stable deduplication identity so repeated deliveries remain idempotent.
- Keeps broker vendor and deployment topology as implementation detail.
- Key sections: 4. Definitions, 5. Normative Requirements, 7. Decision Record, 8. Acceptance Outcome.

RFC-0059: Untrusted Metadata Channels in Agentic SDLC

- **Status:** Living
- **Owner:** Anthesis Core
- **Created:** 2026-04-10
- **Last Updated:** 2026-04-13
- **Related RFCs:** RFC-0003, RFC-0006, RFC-0007, RFC-0009, RFC-0011, RFC-0015, RFC-0027, RFC-0032, RFC-0048

Defines Anthesis doctrine for treating repository and workflow metadata as untrusted input by default when metadata crosses interpretation boundaries.

Summary

- Defines metadata channels as security-relevant inputs, not inert labels.
- Introduces a taint-state lifecycle for metadata that may cross prompts, shells, templates, CLI arguments, logs, and policy engines.
- Requires normalization and validation before interpretation, with explicit boundary handling.
- Clarifies Anthesis enforcement responsibilities and residual risk when metadata leaves its original storage context.
- Keeps boundary-specific runtime mechanics in adjacent control and execution RFCs.
- Key sections: Metadata Trust Model, Taint-State Lifecycle, Boundary Taxonomy, Validation Requirements, Audit and Observability Requirements, Relationship to Other Controls.

RFC-0060: Scope Boundary Governance and Drift Control in Anthesis

- **Status:** Living
- **Owner:** Anthesis Core
- **Created:** 2026-04-19
- **Last Updated:** 2026-04-19
- **Related RFCs:** RFC-0017, RFC-0038, RFC-0054

This RFC defines how Anthesis detects, constrains, escalates, and audits scope expansion during agent-assisted planning and execution.

Summary

- Treats scope creep as a governance problem, not only a planning discipline problem.
- Requires explicit scope boundaries inherited by downstream artifacts.
- Defines validation points before execution and escalation paths when scope expansion is needed.
- Keeps scope changes flowing into governance artifacts rather than sideways through execution.
- Requires drift and scope expansion to remain visible for audit and replay.
- Key sections: 4. Governance Model, 5. Scope Artifact, 6. Control Points Across the Anthesis Loop, 7. Scope Drift Classes.

RFC-0061: Structured Prompt Contracts

- **Status:** Living
- **Owner:** RJ
- **Created:** 2026-04-28
- **Last Updated:** 2026-05-07
- **Related RFCs:** RFC-0005, RFC-0007, RFC-0008, RFC-0009, RFC-0025, RFC-0030, RFC-0034, RFC-0054, RFC-0056, RFC-0057, RFC-0062

Defines Structured Prompt Contracts as versioned, machine-readable role invocation specifications for governed Anthesis executions.

Summary

- Defines Structured Prompt Contracts as versioned, machine-readable role invocation specifications for governed Anthesis executions.
- Establishes SPCs as spec-layer artifacts rather than prompts, RFCs, or execution records.

- Binds role intent, schemas, prompt template references, constraints, invariants, validation expectations, and replay evidence requirements into one governed invocation contract.
- Reinforces that prompt text does not grant authority and that SPC outputs remain subject to Anthesis policy, approvals, and execution boundaries.
- Key sections: 3. Core Doctrine, 8. Contract Model, 10. Normative Requirements, 11. Evidence Envelope Additions, 12. Replay and RER, 13. Security Considerations, 15. Adoption Plan.
- Canonical source: `meristem/rfc_0061_structured_prompt_contracts.md`

RFC-0062: Spartan Prompt Contracts and Token-Aware Execution Profiles

- **Status:** Living
- **Owner:** RJ
- **Created:** 2026-04-28
- **Last Updated:** 2026-05-07
- **Related RFCs:** RFC-0007, RFC-0027, RFC-0034, RFC-0056, RFC-0060, RFC-0061

Defines Spartan Prompt Contracts as governed token-aware execution profiles that compress prompt representation without weakening Anthesis governance semantics.

Summary

- Defines Spartan Prompt Contracts as governed token-aware execution profiles that compress prompt representation without weakening Anthesis governance semantics.
- Treats Spartan prompts as contract-constrained execution profiles rather than a writing style or brevity convention.
- Establishes semantic minimums and compression boundaries so token reduction does not remove scope, authority, provenance, or replay-relevant information.
- Requires structured prompt fields, bounded outputs, and normalization before governed execution.
- Reinforces that compression must preserve governance semantics, replayability, and trust boundaries.
- Key sections: 3. Core Principle, 7. Required Profile Fields, 8. Canonical Shape, 10. Integration with Anthesis, 11. Normalization, 12. Failure Modes, 13. Security Considerations.
- Canonical source: `meristem/rfc_0062_spartan_prompt_contracts_and_token_aware_execution_profiles.md`

RFC-0063: Governance Signals and Signal Lifecycle

- **Status:** Living
- **Owner:** RJ
- **Created:** 2026-04-28
- **Last Updated:** 2026-05-20
- **Related RFCs:** RFC-0011, RFC-0025, RFC-0037, RFC-0054

Defines Governance Signals as the unifying lifecycle for telemetry, feedback, drift, evaluation findings, replay comparisons, policy observations, and evidence-backed system-evolution inputs.

Summary

- Defines Governance Signals as the unifying lifecycle for telemetry, feedback, drift, evaluation findings, replay comparisons, policy observations, and evidence-backed system-evolution inputs.
- Separates raw signals, classified signals, and governed signals to prevent telemetry or feedback from silently becoming governance authority.
- Treats feedback, drift, replay comparisons, and evaluation findings as specializations within one governed signal lifecycle rather than competing governance planes.
- Requires explicit routing, review, promotion, and versioned change application before signals may influence Anthesis behavior.
- Reinforces that signals describe the system while governance decides whether the system changes.
- Key sections: Core Principle, Signal Lifecycle, Signal Envelope, Signal Classes, Relationship to Drift Engineering, Relationship to Replay and Evaluation, Security Considerations, Implementation Status.
- Canonical source: `meristem/rfc_0063_governance_signals_and_signal_lifecycle.md`

RFC-0064: Governed Workflow Composition in Anthesis

- **Status:** Living
- **Owner:** RJ
- **Created:** 2026-05-06
- **Last Updated:** 2026-05-21
- **Related RFCs:** RFC-0003, RFC-0007, RFC-0008, RFC-0009, RFC-0013, RFC-0048, RFC-0054, RFC-0056, RFC-0057, RFC-0059, RFC-0060, RFC-0061, RFC-0063, RFC-0067

Defines how existing Anthesis governance primitives compose into governed workflow execution without introducing a second workflow governance plane.

Summary

- Defines how existing Anthesis governance primitives compose into governed workflow execution without introducing a second workflow governance plane.
- Treats workflows as governed loops composed from existing Anthesis primitives such as routing, scope boundaries, evidence bundles, execution artifacts, and review artifacts.
- Distinguishes advisory workflows from governed workflows and reinforces that exploratory analysis is not equivalent to governed delivery.
- Requires explicit execution and review artifacts, replay linkage, routing visibility, and evidence preservation for governed execution.
- Reinforces that workflow composition must not create hidden governance, routing, approval, or prompt authority.
- Key sections: 4. Core Doctrine, 5. Minimal Workflow Object Model, 6. Workflow Posture, 7. Reference Workflow Composition, 8. Deference Table, 9. Security and Governance Considerations, 10. Decision Record.
- Canonical source: `meristem/rfc_0064_governed_workflow_composition.md`

RFC-0065: Docs Routing and Bounded Context Reads

- **Status:** Living
- **Owner:** RJ
- **Created:** 2026-05-06
- **Last Updated:** 2026-05-07
- **Related RFCs:** RFC-0007, RFC-0017, RFC-0025, RFC-0034, RFC-0036, RFC-0037, RFC-0051, RFC-0060

Defines advisory documentation routing and bounded context-read patterns for repository-aware tools without replacing canonical Anthesis authority.

Summary

- Defines advisory documentation routing and bounded context-read patterns for repository-aware tools without replacing canonical Anthesis authority.
- Introduces lightweight routing artifacts such as entry READMEs, `map.yaml` manifests, domain manifests, and retrieval slices.
- Establishes explicit authority posture between canonical surfaces and advisory documentation layers.
- Reduces token waste, retrieval drift, and accidental cross-domain expansion by encouraging bounded read paths and explicit escalation.
- Reinforces that routing artifacts act as retrieval control surfaces rather than replacement documentation corpora.

- Key sections: 5. Core Principle, 6. Authority Model, 7. Routing Artifact Types, 8. Routing Rules, 10. Anthesis Application, 11. Security and Governance Considerations, 15. Decision Record.
- Canonical source: `meristem/rfc_0065_docs_routing_and_bounded_context_reads.md`

RFC-0066: Conflict Detection and Resolution

- **Status:** Review
- **Owner:** Anthesis Core
- **Created:** 2026-04-28
- **Last Updated:** 2026-05-20
- **Related RFCs:** RFC-0003, RFC-0009, RFC-0014, RFC-0016, RFC-0020, RFC-0023, RFC-0036, RFC-0037, RFC-0038, RFC-0042, RFC-0048, RFC-0051, RFC-0054, RFC-0056, RFC-0060, RFC-0063

Defines conflict as formal artifact incompatibility under governance rules, not agent disagreement, with auditable conflict artifacts and subordinate resolution authority.

Summary

- Defines first-class conflict detection and resolution across human and agent artifacts, with typed conflict artifacts, RFC-0020 phase hooks, and layered integration with governance signals and Phloem/Calyx authority.
- Conflict detectors emit evidence-bearing conflict artifacts; they do not authorize execution.
- Material conflicts route through RFC-0063 signals into Phloem/Calyx or explicit human approval.
- Detection hooks map onto the RFC-0020 workflow phase machine without adding a parallel lifecycle.
- Deterministic structural, resource, and policy checks precede bounded semantic classification.
- Key sections: Normative Inheritance, Conflict Taxonomy, Conflict Artifact, RFC-0020 Phase Mapping, Resolution Strategies, Decision Record.
- Canonical source: `meristem/rfc_0066_conflict_detection_and_resolution.md`

RFC-0067: Recursive Coordination Profiles for Governed Agent Workflows

- **Status:** Review
- **Owner:** RJ
- **Created:** 2026-05-17
- **Last Updated:** 2026-05-20

- **Related RFCs:** RFC-0009, RFC-0013, RFC-0034, RFC-0047, RFC-0048, RFC-0054, RFC-0059, RFC-0060, RFC-0061, RFC-0063, RFC-0064

Profiles recursive coordination atop RFC-0009, RFC-0013, RFC-0054, and RFC-0064 with explicit depth, evidence, and replay propagation boundaries.

Summary

- Defines Recursive Coordination Profiles for bounded recursive, dynamic, latent, or externally orchestrated agent coordination without a second workflow plane.
- RCPs are metadata and policy overlays, not a replacement orchestration engine.
- Supports declared static and bounded dynamic expansion in initial scope.
- Preserves Phloem/Calyx approval and RFC-0020 phase subordination.
- Opaque external recursive executors require boundary-level evidence only.
- Key sections: Recursive Coordination Profile Model, Depth and Budget Constraints, Replay and Evidence Propagation, Relationship to RFC-0064, Decision Record.
- Canonical source: `meristem/rfc_0067_recursive_coordination_profiles_for_governed_agent_wor`

RFC-0068: Temporal Governance for Interaction-Derived Execution

- **Status:** Review
- **Owner:** RJ
- **Created:** 2026-05-18
- **Last Updated:** 2026-05-20
- **Related RFCs:** RFC-0003, RFC-0007, RFC-0009, RFC-0030, RFC-0034, RFC-0047, RFC-0048, RFC-0054, RFC-0056, RFC-0057, RFC-0059, RFC-0060, RFC-0061, RFC-0063, RFC-0064, RFC-0067

Interaction is not execution; continuous collaboration must stabilize intent before a governed execution candidate may form under existing Anthesis law.

Summary

- Defines temporal governance for interaction-derived execution, including interaction epochs, intent stabilization, authority transitions, and interaction evidence extensions without equating interaction with execution.
- Interaction epochs are temporal governance windows, not workflow phases.
- Intent fragments are observations, not authority, until stabilization criteria pass.

- Extends RFC-0054 replay evidence and RFC-0063 signal expectations for interaction streams.
- Subordinate to RFC-0047 session governance and RFC-0020 execution phases.
- Key sections: Core Doctrine, Interaction Authority Model, Intent Stabilization Floor, Interaction Evidence Extension to RFC-0054, Decision Record.
- Canonical source: `meristem/rfc_0068_temporal_governance_for_interaction_derived_execution`.

RFC-0099: Anthesis v1 Unfreeze & Amendment Rules

- **Status:** Living
- **Owner:** Anthesis Core
- **Created:** 2025-12-19
- **Last Updated:** 2026-05-07
- **Related RFCs:** RFC-0000, RFC-0003, RFC-0017, RFC-0020, RFC-0023, RFC-0030, RFC-0033

This RFC transitions Anthesis v1 from Frozen to Active (Amendments Allowed).

Summary

- This RFC transitions Anthesis v1 from Frozen to Active (Amendments Allowed).
- Key sections: 6. Constitutional and Extension Boundaries, 9. RFC Hygiene and Normalization (Normative), 10. Emergency Security Amendments, 11. Precedence and Conflict Handling, 12. Lifecycle Controls.

Calyx: Policy and Approval Governance

Calyx is Anthesis' **policy and approval governance system**. It defines how changes are classified by risk, which execution candidates require human approval, and how the system maintains accountability over agentic decisions.

Purpose

Calyx exists to answer:

Is this action allowed? Who must approve it? Can we prove they did?

Calyx is the **policy decision layer** that ensures agents operate only within explicitly defined boundaries, with transparent audit trails. Phloem records and enforces the resulting control-plane decisions.

Core Concepts

Bloom Classes Bloom Classes are **risk tiers** assigned to artifacts and changes:

Class	Label	Definition	Approval Required
0	Read-Only	Documentation, analysis	No
1	Low Risk	Prompts, tests, minor refactors	No
2	Medium Risk	Application code, workflows	Yes (reviewer)
3	High Risk	Infrastructure, CI/CD, security	Yes (reviewer)
4	Critical	Production deploys, secrets, access	Yes (admin)

Bloom Classes are determined by:

- **File path** (mapped in `bloom-classes.yaml`)
- **Change type** (refactor vs new feature vs delete)
- **Scope** (single file vs repository-wide)

Approval Policy Approval policies **declare when and how approvals are required**.

Policies are evaluated for every governed execution candidate:

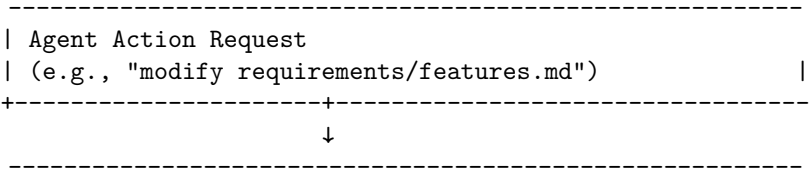
1. **Classify** the change (Bloom Class)
2. **Bind** scope, artifact identity, authority, and evidence requirements
3. **Match** against rules
4. **Require** approvals if rule specifies
5. **Halt** until approvals collected
6. **Audit** decision, approver, rationale, and affected artifacts

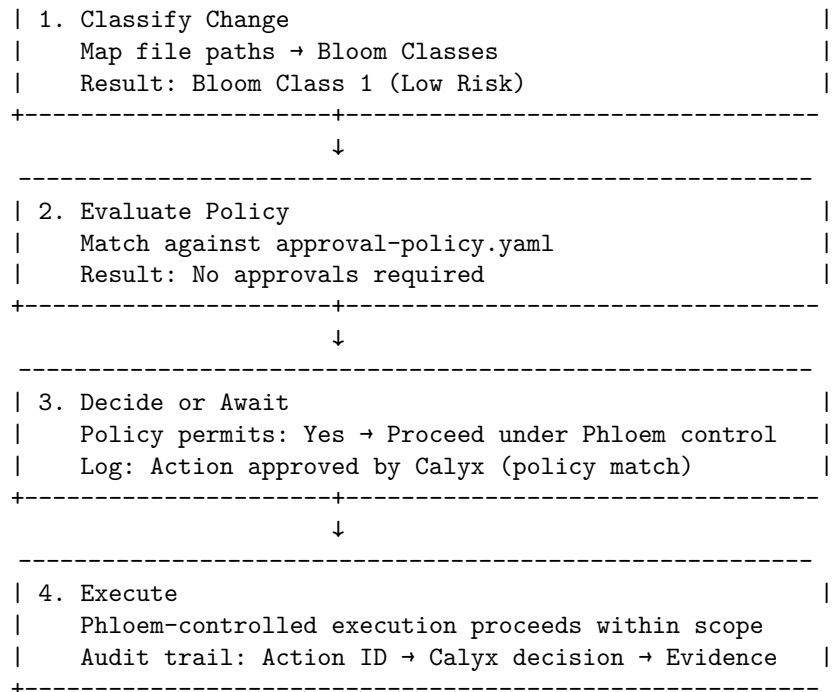
Approvals are not ambient. RFC-0030 requires approvals to map to exact artifacts, usually by path plus commit identity. An approval for one proposal, execution, or artifact must not silently bless unrelated changes.

Conversational acknowledgement is not approval. RFC-0068 requires interaction-derived execution to cross an intent stabilization boundary before approval can authorize mutation.

Execution Flow

When an agent action is submitted to Phloem:





Policy Evaluation Outcomes

Outcome	Description	Next Step
Auto-Approve	No approvals required	Execute immediately
Await Approval	Human approval needed	Halt; wait for approver
Reject	Policy forbids action	Fail; log reason
Escalate	Route to higher authority	Forward to admin channel
Signal	Advisory governance concern detected	Record signal; evaluate policy impact

Signal existence alone is not approval, rejection, or execution authority. RFC-0063 signals feed governance evaluation; Calyx or Phloem decisions determine whether they block, escalate, or only annotate an execution.

Conflict and Interaction Handling

Calyx decisions may consume conflict artifacts and interaction evidence:

- RFC-0066 conflict artifacts describe incompatibility between plans, executions, reviews, policies, or resources.

- Conflict detectors do not authorize execution. Material conflicts route into Calyx, Phloem, reviewer, or human approval boundaries.
- RFC-0068 interaction evidence helps distinguish unstable intent from an execution candidate.
- Approval language captured during interaction must be preserved as evidence, then evaluated under the applicable policy before mutation.

Approval Channels

Approvals can be collected from:

1. **GitHub PR Reviews** - Reviewer approves PR
2. **CLI Confirmation** - Human confirms in terminal
3. **n8n Workflows** - Approval action in workflow
4. **MCP API** - Programmatic approval submission

All approvals are **logged immutably** with:

- Approver identity
- Timestamp
- Action ID
- Decision rationale (if provided)

Example Scenarios

Scenario 1: Low-Risk Change (Auto-Approve)

Agent: "Update prompts/system-message.md"
 File: prompts/system-message.md
 Bloom Class: 1 (Low Risk)
 Policy Match: default_low_risk
 Approvals Required: []
 Result: Auto-approved → Execute

Scenario 2: Medium-Risk Change (Require Review)

Agent: "Refactor src/lib/domain/agent.ts"
 File: src/lib/domain/agent.ts
 Bloom Class: 2 (Medium Risk)
 Policy Match: default_medium_plus
 Approvals Required: [reviewer]
 Result: Awaiting Approval → Notify reviewers

Scenario 3: Critical Change (Require Admin)

Agent: "Update .env with new API key"
 File: secrets/.env.production
 Bloom Class: 4 (Critical)
 Policy Match: critical_secrets

Approvals Required: [admin]

Result: Admin approval only → Escalate to admin channel

Audit Logs Every Calyx decision is logged to:

- **MCP Event Log** - Structured JSON
- **Evidence Records** - Approval, artifact, policy, and execution linkage
- **External SIEM** - Via audit sink (if configured)

This ensures a **verifiable audit trail** for compliance and review.

Reference

Related RFCs

- RFC-0003: Approval Policy Engine - Calyx policy and approval model
- RFC-0014: Risk Classification and Bloom Classes - Risk tier inputs
- RFC-0030: Approval-Artifact Mapping - Artifact-scoped approval semantics
- RFC-0048: Phloem Policy Evaluation and Decision Contract - Phloem decision and attestation boundaries
- RFC-0063: Governance Signals and Signal Lifecycle - Advisory signal handling
- RFC-0066: Conflict Detection and Resolution - Conflict artifacts and routing
- RFC-0068: Temporal Governance for Interaction-Derived Execution - Intent stabilization and approval boundaries

Inflorescence: Multi-Agent Coordination

Inflorescence is Anthesis' **multi-agent coordination system**. It defines how multiple agents collaborate within governed workflow graphs, enabling complex, observable, replayable, and safe agent-to-agent coordination.

Purpose

Inflorescence answers:

*How can multiple agents work together on a single problem safely?
Who depends on whom? What happens when something fails?*

Inflorescence is the coordination layer that turns individual governed executions into a bounded workflow. It does not create a second governance plane; Calyx and Phloem policy, approval, execution, and evidence rules still apply.

Core Concepts

Coordination Graphs Coordination graphs are **directed acyclic graphs (DAGs)** where:

- **Nodes** represent agent executions
- **Edges** represent data flow or approval dependencies
- **Workflows** describe task sequences, fan-out/fan-in, and failure strategies

Graphs are declarative, versioned in Git, and enforced by Anthesis at runtime.

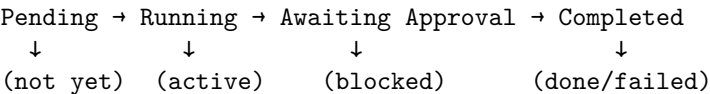
Each graph belongs to one repository in v1. Cross-repository coordination is signal-only unless future governance explicitly expands that boundary.

Coordination Patterns Inflorescence supports standard workflow patterns:

Pattern	Description	Use Case
Sequential	$A \rightarrow B \rightarrow C$	Linear workflows
Fan-out	One agent spawns many	Parallel analysis
Fan-in	Multiple agents $\rightarrow$ aggregator	Consolidate results
Map/Reduce	Partition $\rightarrow$ analyze $\rightarrow$ synthesize	Large-scale tasks
Supervisor/Worker	Controller delegates scoped tasks	Distributed execution
Conditional	Branch on intermediate results	Decision trees
Bounded recursive	Policy-bounded graph expansion	Recursive planning, review, synthesis

Open-ended recursion remains disallowed by default. RFC-0067 permits recursive coordination only through explicit Recursive Coordination Profiles, with depth, node, budget, approval, evidence, and replay limits.

Execution States Each node in a graph transitions through states:



Workflow Format

Inflorescence workflows are **YAML files** that define nodes, edges, and execution parameters.

Example 1: Simple Sequential Three agents run in sequence:

1. Agent A: Gather Requirements
2. Agent B: Validate Requirements
3. Agent C: Finalize Documentation

Example 2: Fan-Out / Fan-In One trigger spawns multiple parallel agents, then aggregate.

1. Read Specification file
2. Fanout: Three parallel agents analyze different sections
 - Agent A: Analyze Performance
 - Agent B: Analyze Security
 - Agent C: Analyze Usability
3. Fan-in: Aggregator agent synthesizes findings into a report

Example 3: With Approval Gate Workflow with human approval checkpoint:

1. Agent A: Deployment Planner
2. Agent B: Deployment Review Requires Approval
3. Agent C: Execute Deployment (only if approved)

Example 4: Failure Handling Workflow that continues if some nodes fail

1. Agent A: File Analysis (must succeed)
2. Agent B: File Analysis (must succeed)
3. Agent C: File Analysis with leniency (if it fails, log and continue)

Example 5: Bounded Recursive Review Workflow with a declared recursive profile:

1. Planner decomposes RFC review into bounded subreviews
2. Reviewers run under max depth, max node, and budget limits
3. Aggregator records evidence from all governed delivery-path branches
4. Human or policy approval gates any escalation or promotion

Validation & Safety

Inflorescence enforces:

- **DAG constraint:** No cycles allowed
- **Acyclic graphs:** Prevent infinite loops
- **Execution limits:** Max depth, max nodes per execution
- **Timeouts:** Per-node and global workflow timeouts
- **No hidden shared mutable state:** Agents communicate through declared edges and evidence
- **Immutable history:** All executions logged and replayed within declared evidence limits
- **Recursive profile limits:** Dynamic expansion requires explicit policy allowance
- **Conflict handling:** Incompatible plans, resources, reviews, or policies emit conflict artifacts

- **Interaction boundaries:** Continuous interaction must stabilize before becoming execution

Agents Agents are nodes in workflows. They:

- Receive inputs via templated substitution
- Produce JSON output
- Can specify approval requirements
- Respect Calyx policies
- Do not inherit authority from parent nodes unless policy explicitly permits it
- Preserve lineage and evidence links for replay

Inflorescence + Calyx Workflows are subject to Calyx policies:

Workflow → Node Execution → Calyx Classification → Approval Gate

Child execution approval is not automatic. Recursive expansion, high-risk branches, conflict resolution, and artifact promotion may require separate approval artifacts.

Inflorescence + Temporal Governance Interaction epochs are not workflow phases. A workflow may contain interaction epochs, and interaction may precede workflow creation, but RFC-0068 requires intent stabilization before interaction-derived output becomes a governed execution candidate.

Monitoring & Observability

Every state transition emits an event that is queryable via MCP.

Metrics

- Workflow execution time
- Node success/failure rates
- Approval wait time
- Data flow volume
- Approval decision distribution
- Conflict rate and resolution latency
- Recursive depth, node count, budget use, and replay classification

Best Practices

1. **Keep workflows simple** - Fewer than 10 nodes per workflow
2. **Name nodes clearly** - Use descriptive, actionable IDs
3. **Document inputs/outputs** - Specify data contracts
4. **Set appropriate timeouts** - Prevent hung executions
5. **Plan failure strategies** - Decide at design time
6. **Version workflows** - Increment version on changes

7. **Test locally** - Use workflow simulator before deploying
8. **Monitor execution** - Watch approval and failure metrics
9. **Iterate** - Refine workflows based on real-world usage

Reference

Related RFCs

- RFC-0013: Multi-Agent Coordination - Coordination patterns
- RFC-0009: Execution Semantics - Execution states
- RFC-0005: Agent Schema - Agent definitions
- RFC-0054: Execution Evidence and Replay Envelope - Replay evidence
- RFC-0064: Governed Workflow Composition - Workflow composition boundaries
- RFC-0066: Conflict Detection and Resolution - Conflict artifacts
- RFC-0067: Recursive Coordination Profiles - Recursive coordination bounds
- RFC-0068: Temporal Governance for Interaction-Derived Execution - Interaction and stabilization boundaries

Services: The Execution Layer of Anthesis

Microservices that power Anthesis: Phloem (MCP server & orchestration) and Xylem (worker execution service).

Services execute governed work; they do not create independent governance authority. Calyx policy, Phloem decisions, execution evidence, and Meristem RFCs remain the control boundaries for approvals, conflicts, replay, recursive coordination, and interaction-derived execution.

Service Architecture

```
Services (Tier 3: Execution)
|
+-- Phloem (Orchestration & API)
|   +-- MCP Server implementation
|   +-- Request routing and validation
|   +-- Policy integration (Calyx)
|   +-- Proposal management
|   +-- Conflict and governance signal routing
|   +-- Outbox execution queue
|   +-- State management
|   +-- Evidence and audit logging
|
+-- Xylem (Worker Execution)
|   +-- Message queue consumer
|   +-- Agent execution
```

```

|   +--- Context assembly
|   +--- Governed workflow execution
|   +--- Output validation
|   +--- Result collection
|   +--- Error handling
|   +--- Queue initialization (scripts/)
|
+-- Xylem (Embedded Execution)
    +--- Local execution mode for CLI
    +--- Same execution logic as worker
    +--- Runs in-process with CLI for low-latency operations

```

Service Communication

Request Flow

1. Client → Phloem: Request
2. Phloem → Calyx: Classification, scope, and approval evaluation
3. Phloem: Conflict/signal/evidence checks
4. Phloem → Xylem: Dispatch
5. Xylem → Phloem: Claim
6. Xylem → Phloem: Execute/report evidence
7. Phloem: Record decision, state, and replay evidence
8. Response to Client

Service SLOs

Target service level objectives:

Metric	Target
Availability	99.9% (9h/month downtime)
Latency (p50)	< 500ms
Latency (p99)	< 5s
Error Rate	< 0.1%
Queue Latency	< 1s (p95)

Phloem Service: The Anthesis MCP Server and Orchestration Hub

Phloem is the Anthesis MCP server implementation. It exposes the orchestration API, enforces policy via Calyx, and records execution/audit data for governed workflows.

Purpose: Central orchestration hub, MCP server, policy decision enforcement, and evidence recording

Responsibilities:

- Receive and parse incoming requests
- Classify requests with Calyx
- Bind scope, artifact identity, and approval requirements
- Route governance signals and conflict artifacts
- Route work to executors
- Aggregate and return results
- Record audit and replay evidence
- Handle approvals and callbacks

Phloem does not replace Calyx approval policy or Meristem governance. It records and enforces decisions at the control-plane boundary, including conflict, recursive workflow, and interaction-derived execution constraints.

Configuration

Runtime Posture The runtime posture is modeled as four axes:

- `operating_mode`: `offline` | `local` | `remote`
- `execution_topology`: `embedded` | `worker`
- `relay_mode`: `embedded` | `process`
- `deployment_scope`: `single` | `distributed`

Canonical runtime profiles and configuration overlays can be defined as combinations of these axes. See Configuration for details on how to configure these profiles and overlays.

Queue Backend Configuration Phloem uses a vendor-agnostic queue abstraction layer. Configure via environment variables.

Xylem Service: The Anthesis Execution Agent and Queue Consumer

Xylem is the execution agent for Anthesis. It consumes dispatch messages from the Xylem message queue, claims executions from the Phloem MCP server, and relays dispatch notifications back to the API.

Xylem has two operating modes depending on runtime configuration:

1. **Queue Consumer Mode** (default) - Subscribes to configured queue backend, consumes dispatch messages, claims executions, and relays notifications.
2. **Embedded Mode** - Runs as a standalone process without queue consumption, useful for testing or environments without queue backends.

Xylem operates with Phloem (MCP server) to claim and execute tasks/workflows and relay notifications. It also supports health monitoring, reporting, and graceful shutdown.

Xylem executes only within the scope Phloem authorizes. It does not grant approval, expand authority, or bypass conflict, replay, recursive coordination, or interaction-derived execution constraints.

Responsibilities:

- Subscribe and initialize work queues
- Load execution context
- Execute agents with scoped access
- Validate output
- Preserve execution lineage and replay evidence
- Handle errors gracefully
- Report results back
- Health monitoring

Worker Types:

- **Agent Worker** - Executes agents
- **Tool Worker** - Executes external tools
- **Approval Worker** - Handles approval callbacks
- **Notification Worker** - Sends notifications

Execution Flow

1. Worker receives dispatch message from queue backend
2. Extracts `execution_id` from payload
3. Claims execution: `POST /v1/executions/{id}/claim`
4. Executes agent or workflow with context from MCP under authorized scope
5. Relays dispatch: `POST /v1/executions/{id}/dispatch`
6. Acknowledges message to queue backend
7. Reports status to provenance queue (if enabled)
8. Handles errors and retries according to policy

Scaling:

- Horizontal scaling (add more workers)
- Different worker pools for different task types
- Prefetching and batching for efficiency
- Auto-restart on failure
- Graceful shutdown on signals

Tooling: Building, Testing, and Deploying Anthesis

Development tools and utilities for Anthesis: CLI, AI client library, templating system, and test infrastructure. Everything needed to build, test, and deploy the platform.

Included Tools

1. CLI (Command-Line Interface) Anthesis CLI is the local control surface for governed workflows, providing scriptable commands that mirror MCP operations while enforcing Anthesis policy and plugin contracts.

2. Embedded Runtime Anthesis Embedded Runtime is a lightweight execution environment for running agents and workflows without a worker deployment. It provides a direct API to the Phloem MCP server and manages context in-memory, making it ideal for testing, development, and environments without queue backends.

3. AI Client Library `ai_client` is a shared Python library providing OpenAI-compatible client helpers for AI runtime integration. It manages model discovery, configuration loading, context, and chat completion requests across Anthesis services and plugins. It supports multiple backends (Ollama, LM Studio, etc.) and provides a consistent interface for AI interactions.

4. Observability & Logging Standardized logging and observability utilities for consistent instrumentation across services and agents. Provides structured logging, OpenTelemetry integration, and centralized configuration for log levels and formats. This ensures that all components of Anthesis emit consistent, queryable logs that can be easily integrated with monitoring and alerting systems.

5. Message Queue Abstraction Vendor-agnostic message queue abstraction layer supporting RabbitMQ, Redis Streams, Apache Kafka, and AWS SQS. Provides a consistent interface for enqueueing and consuming messages, allowing Phloem and Xylem to operate with different queue backends based on deployment needs. Configuration is handled via environment variables, enabling flexibility without code changes.

6. Templates Shared Jinja2 templates for Anthesis tooling, distribution packaging, and test scaffolding. Templates ensure consistency in generated artifacts, such as agent definitions, workflow graphs, and plugin manifests. They can be extended by plugins to provide domain-specific templates for various use cases.

Template Types:

- **Workflow templates** - Assemble execution context for agents
- **Artifact templates** - Format changes as commits or diffs
- **Context templates** - Build agent input from previous outputs
- **Output templates** - Format agent output for downstream use

7. Test Infrastructure Mainly focused on integration testing between control surfaces, data, and services. Tests are split into **unit** and **integration**. Integration testing happens through services and the CLI in docker compose. Coverage is tracked and CI/CD pipelines enforce minimum thresholds.

Tools Overview

Tool	Purpose	Status
CLI	Command-line interface	Active
Embedded Runtime	Local execution environment	Active
ai_client	AI runtime library	Active
templates	Jinja2 templating	Active
message_queue	Message queue abstraction	Active
observability	Logging and instrumentation	Active
tests	Test infrastructure	Active

Anthesis CLI: The Local Control Surface for Governed Workflows

Anthesis CLI is the local control surface for governed workflows, providing scriptable commands that mirror MCP operations while enforcing Anthesis policy and plugin contracts.

Features

- Sessions for invariant and overlay prompts, context, and configuration
- Plugin command execution with policy enforcement
- Execution of agents and workflows
- Agent management
- Model governance and configuration
- RFC management
- AI client library for model interactions
- Observability utilities for consistent logging and instrumentation
- System runtime profiles for different environments (local, staging, production)
- Tool for integration testing

Plugin registry loading

The CLI loads plugins from a registry configured in user or project scope. The registry defines available plugins, their sources (local or release), and configuration.

Configuration hierarchy

Anthesis CLI configuration is loaded from multiple sources with the following precedence (highest to lowest):

1. CLI arguments
2. Environment variables
3. User config file (`~/.config/anthesis/config.yaml`)
4. Project config file (`anthesis/config.yaml`)

Session Profiles Session profiles allow prompt configurations, context, and artifacts to be defined.

Use `anthesis session profiles` to inspect the configured aliases and see how they map to SDLC or product-lifecycle families:

```
anthesis session profiles
anthesis session profiles --aliases-only
anthesis session profiles --explain sdlc-analysis
anthesis session profiles --explain sdlc-release
anthesis session profiles --explain product-discover
anthesis session profiles --explain product-validate
```

Runtime Profiles Runtime profiles define the configuration to access the various services and resources in different environments (local, staging, production). This includes API endpoints, authentication, model configurations, and plugin settings.

Configuration: Design and Documentation

Anthesis configuration is designed to be hierarchical, environment-aware, and security-conscious. It supports multiple layers of configuration (project, user, environment) with clear precedence rules. Sensitive information is handled securely with support for secrets management and encryption. Configuration can be dynamically reloaded to reflect changes without restarting services. This design allows for flexible and secure management of Anthesis across different environments and use cases.

Configuration Layers

1. **Project Configuration** - `anthesis/config.yaml` (optional, version-controlled)
2. **User Configuration** - `~/.config/anthesis/config.yaml` (optional, user-specific)
3. **Environment Variables** - `ANTHESIS_*` (optional, overrides file configs)
4. **CLI Arguments** - `--config` and other flags (optional, highest precedence)

Security Considerations

- Sensitive information (API keys, credentials) should be stored in environment variables or secrets management systems, not in version-controlled files.
- Configuration files should have appropriate permissions to restrict access.
- Support for encrypted configuration values can be implemented for added security.

Dynamic Reloading

Anthesis services can be designed to watch configuration files for changes and reload them dynamically without requiring a restart. This allows for seamless updates to configuration in production environments.

Profiles & Overlays

Configuration profiles can be defined for different environments (local, staging, production) with specific settings. Overlays can be used to apply environment-specific overrides on top of base configurations. This allows for flexible management of configuration across different deployment targets.

Architecture: Decision Records and Design Notes

Architecture decision records, system design notes, and technical diagrams. This section captures the architectural decisions, design patterns, and system structure of Anthesis. It includes documentation on component interactions, data flows, integration patterns, and non-functional requirements such as performance, reliability, and security. The architecture documentation serves as a reference for developers and stakeholders to understand the technical design and rationale behind key decisions in the Anthesis system.

Scope

- ADRs and architectural trade-offs
- System diagrams and component boundaries
- Integration patterns and data flows
- Non-functional requirements (performance, reliability, security)

Conventions

- Use ADRs for decisions that affect long-term structure.
- Link to related RFCs and specs.
- Keep diagrams source-backed (Mermaid, draw.io, etc.).

Requirements: Defining the "What" of Anthesis

Anthesis requirements define the necessary conditions, capabilities, and constraints that must be met for the successful implementation and operation of the Anthesis system. These requirements encompass functional aspects such as agent behavior, workflow execution, plugin interactions, and session management, as well as non-functional aspects like performance, scalability, security, and maintainability. By establishing clear requirements, stakeholders can ensure that the development process is aligned with the intended goals and use cases of Anthesis, facilitating effective design decisions and prioritization throughout the project lifecycle.

Requirement Categories

1. **Functional Requirements** - Define specific behaviors, features, and interactions that the Anthesis system must support, such as agent capabilities, workflow orchestration, plugin functionality, and session management.
2. **Non-Functional Requirements** - Specify performance, scalability, security, reliability, and maintainability criteria that the Anthesis system must meet to ensure a robust and efficient operation.
3. **Governance Requirements** - Outline policies, rules, and constraints for agent behavior, workflow execution, and plugin interactions to ensure responsible and ethical use of AI capabilities.
4. **Integration Requirements** - Detail the necessary interfaces, protocols, and data formats for seamless integration between Anthesis components (MCP, Phloem, Xylem) and external systems or services.
5. **Testing & Validation Requirements** - Define the criteria and methodologies for testing the functionality, performance, and security of the Anthesis system to ensure it meets the specified requirements.
6. **Documentation Requirements** - Specify the necessary documentation for users, developers, and administrators to effectively utilize, extend, and maintain the Anthesis system.

Specifications: The Technical Blueprints of Anthesis

Anthesis specifications define the expected technical structure, behavior, and interfaces for various components of the Anthesis system. These specifications serve as a blueprint for development, ensuring consistency, interoperability, and adherence to design principles across different modules and plugins. Specifications cover areas such as configuration formats, plugin interfaces, API contracts, data models, and communication protocols. By following these specifications, developers can create compatible components that integrate seamlessly within the Anthesis ecosystem, facilitating extensibility and maintainability.

Specification Categories

1. **Governance Specifications** - Define policies, rules, and constraints for agent behavior, workflow execution, and plugin interactions.
2. **Plugin Specifications** - Outline the expected interfaces, lifecycle, and capabilities for plugins within the Anthesis system.
3. **Session Specifications** - Describe the structure and management of sessions, including context, prompts, and artifacts.
4. **Runtime Specifications** - Detail the expected behavior and interfaces for runtime components such as the MCP server and worker.
5. **API Specifications** - Define the endpoints, request/response formats, and authentication for Anthesis APIs.
6. **Data Model Specifications** - Describe the structure and relationships of data entities used across the system, such as agents, workflows, executions, and artifacts.
7. **Communication Protocol Specifications** - Outline the expected message formats, queues, and protocols for communication between components like Phloem and Xylem.

Plugins: Extending Capabilities with Governance

Plugins extend platform capabilities while preserving governance, policy enforcement, and auditability. They add agents, tools, integrations, and workflows without bypassing Calyx or expanding authority.

Philosophy

Plugins follow a core principle:

Plugins extend capability, never authority.

- Plugins cannot bypass Calyx policies
- Plugins cannot modify approval requirements
- Plugins cannot treat signals, chat acknowledgements, or generated outputs as approval
- Plugins cannot access privileged operations
- Plugins are **fully auditable** and **replayable**
- Plugins have a manifest and an execution model
- Plugins are signed with certificates auditable

Plugins that emit workflows, tool calls, or generated artifacts remain subject to governed workflow composition, conflict detection, recursive coordination profiles, and interaction-derived execution boundaries.

Types

Anthesis supports four plugin categories, each with specific constraints and use cases:

- Agent plugins
- Tool plugins
- Integration plugins
- UI plugins

Document Publisher Plugin: Building Anthesis Documentation from Markdown

This plugin builds Anthesis documentation artifacts from Markdown sources. It converts Markdown files into HTML and PDF formats, integrating with the MCP for artifact storage and retrieval. The plugin is configurable, allowing users to specify source and output paths, as well as custom templates and styling for the generated documentation.

Git Plugin: Local Git Utilities for Anthesis

Local-only git utilities for Anthesis. Provides commit message generation, diff analysis, and integration with the MCP for commit and diff-based workflows. This plugin is designed to assist agents in understanding code changes, generating meaningful commit messages, and analyzing diffs to inform decision-making processes. It can be used in workflows that involve code review, automated commits, or any scenario where understanding code changes is beneficial.

Task Plugin: Local Task Management and Governance Tracking

Task plugin for intended for local task management and governance tracking. This plugin provides capabilities for creating, assigning, and tracking tasks within the Anthesis system, integrating with the MCP for task execution and notifications. It also includes features for governance tracking and reporting on task outcomes, ensuring that all tasks are executed in accordance with defined policies and that their results are properly documented and accessible for review.

CSSLP Analysis Summary

This document is a concise Anthesis-facing summary of the broader CSSLP security assessment. It is intended to orient Anthesis team members to the purpose, scope, and main themes of the CSSLP analysis, and to connect it to relevant RFCs and documentation.

It is explanatory only. Normative behavior remains defined by the Charter and the RFCs in `meristem/`.

Purpose

The CSSLP analysis is used to evaluate Anthesis across the secure software lifecycle, identify control gaps, and prioritize remediation work in a way that fits Anthesis governance rather than bypassing it.

Assessment Position

The current CSSLP assessment indicates:

- a strong baseline in governance, traceability, and auditability
- meaningful gaps in secrets management, dependency management, and validation hardening
- a need to keep security controls tied to existing Anthesis policy and evidence surfaces

How Anthesis Uses CSSLP

Anthesis uses CSSLP as:

- an analysis and review framework
- a source of security improvement priorities
- a way to organize lifecycle-wide security expectations

Anthesis does not use CSSLP as:

- a replacement for RFC governance
- a separate execution authority
- a standalone compliance engine

CSSLP Principles Applied to Anthesis

Anthesis applies CSSLP as a lifecycle-wide security discipline rather than as a single review step. In practical terms, that means:

- security requirements should be explicit, traceable, and tied to governed artifacts
- secure design decisions should be captured in RFCs and architecture records
- implementation controls should combine deterministic validation with governed review
- supply-chain integrity should be treated as part of the SDLC, not a separate concern
- testing and verification should produce attributable evidence rather than informal confidence
- deployment and operational transitions should preserve security posture and auditability
- vulnerability response should feed back into policy, design, and future change control

For Anthesis specifically, these principles map to:

- Charter and RFCs as the canonical design and security intent layer
- Calyx and Phloem as the security gate and policy decision layer
- Xylem, CI, and admitted tools as the validation and evidence-producing layer
- evidence export, provenance, and repository verification as the auditability layer
- human approval and governance review as the final authority layer

Main Themes From the Assessment

Secure Design

- Security should remain part of architectural intent, not a post-build review.
- Trust boundaries, threat models, and security architecture should be documented explicitly.

Secure Requirements

- Security requirements should be traceable to RFCs, implementations, and tests.
- Compliance mappings may be useful, but should remain secondary to concrete Anthesis controls.

Implementation and Validation

- Validation should combine deterministic checks with admitted agent review.
- Security-relevant changes should produce attributable evidence.
- Supply-chain visibility and artifact integrity need to remain first-class concerns.

Response and Improvement

- Vulnerability handling should lead to governed remediation work.
- Repeated security issues should feed back into policy, RFC, or design updates.

Relationship to Current RFCs

The CSSLP analysis aligns most directly with:

- `meristem/rfc_0003_approval_policy_engine.md`
- `meristem/rfc_0010_ci_cd_integration_model.md`
- `meristem/rfc_0023_audit_compliance_evidence_export.md`
- `meristem/rfc_0032_external_agent_tool_admission.md`
- `meristem/rfc_0044_gittuf_git_native_policy_verification_substrate.md`

- `meristem/rfc_0046_ssdf_integration_and_enforcement_model.md`

Summary

The CSSLP analysis is best treated as a structured security review lens over Anthesis, helping convert broad secure-lifecycle concerns into governed, traceable, evidence-backed improvements.

Governed SDLC Security Baseline

This document summarizes the current Anthesis security baseline for governed software delivery. It is explanatory only. The canonical normative sources remain the Charter and the RFCs under `meristem/`.

Purpose

Anthesis treats security as a governed property of the SDLC, not a separate after-the-fact review track. The security baseline is built from four layers:

- governance and approval controls
- execution and validation controls
- Git-native verification controls
- audit and evidence controls

Core Security Posture

- Humans retain final authority over policy, approvals, and trust changes.
- Agents may assist with review and validation, but do not become security authorities.
- Policy gates fail closed when required security evidence is missing.
- Security-relevant changes must remain attributable, reviewable, and exportable as evidence.

Current Control Surfaces

Governance and Approval

- Calyx defines policy and approval requirements.
- Phloem evaluates whether governed actions may proceed.
- Sessions, proposals, approvals, and evidence remain distinct control objects.
- Tracks are coordination-only artifacts and do not create execution permission.

Validation and Enforcement

- Xylem executes or coordinates scans, validation jobs, and supporting security workflows.

- Security agents may review implementation, dependency posture, and SBOM-related outputs.
- Deterministic checks remain authoritative over agent judgment.

Git-Native Verification

- gittuf is used as the Git-native policy and verification substrate for protected refs and protected paths.
- Anthesis governance remains authoritative for semantic decisions; gittuf verifies repository policy compliance.
- Approval bridging links Anthesis-approved decisions to repository verification artifacts.

Evidence and Audit

- Security decisions must be attributable to a human, service, or admitted tool identity.
- Evidence must remain append-only, contextual, and exportable.
- Verification, scan results, approval linkage, and policy decisions together form the reviewable security evidence chain.

Relevant RFCs

- `meristem/rfc_0003_approval_policy_engine.md`
- `meristem/rfc_0009_agent_execution_semantics.md`
- `meristem/rfc_0010_ci_cd_integration_model.md`
- `meristem/rfc_0020_state_machine_formal_specification.md`
- `meristem/rfc_0023_audit_compliance_evidence_export.md`
- `meristem/rfc_0032_external_agent_tool_admission.md`
- `meristem/rfc_0044_gittuf_git_native_policy_verification_substrate.md`
- `meristem/rfc_0045_tracks_as_governed_coordination_artifacts.md`
- `meristem/rfc_0046_ssdf_integration_and_enforcement_model.md`

SAFE-MCP: Structured Security Analysis for MCP-Based Systems in Anthesis

Anthesis adopts SAFE-MCP as a security analysis framework to structure threat modeling, control verification, and detection planning for MCP-based systems. This document describes how SAFE-MCP is used across Anthesis components to maintain a consistent security posture without altering governance or implementation.

Background and Rationale

Anthesis is an agentic SDLC platform with a multi-surface attack profile: MCP server endpoints, CLI clients, multi-agent coordination, plugins, and federated integrations. SAFE-MCP provides a standardized taxonomy of MCP-specific

threats and mitigations. Anthesis uses this taxonomy to align security controls with clear evidence and to avoid ad hoc, inconsistent security reviews.

Scope of Application

SAFE-MCP is applied across the full stack:

- MCP server boundary (Phloem): authentication, authorization, tool exposure, and request validation.
- Client and CLI flows: prompt injection risks, unsafe tool invocation, and approval gates.
- Plugin ecosystem: tool poisoning and supply chain style risks in manifests and install paths.
- Policy and risk tiers: approval policy enforcement, least privilege, and bloom class constraints.
- Observability and audit: detection coverage, provenance logging, and audit trails.
- Federation and integrations: trust boundaries and cross-system access constraints.

Method of Use

Anthesis uses SAFE-MCP in three structured modes:

1. Threat Modeling: Map SAFE-MCP techniques to Anthesis surfaces, identify gaps, and document assumptions.
2. Control Verification: Validate that approvals, allowlists, and enforcement points exist and are auditable.
3. Test Design: Convert techniques into adversarial test cases for MCP routes, tool calls, and plugin workflows.

Outputs and Evidence

The primary artifact is the SAFE-MCP coverage matrix in `docs/security/safe-mcp-analysis.md`. It documents:

- Threat class to component mappings
- Expected controls and detections
- Evidence locations in the repository
- Status labels to track coverage and gaps

Governance Alignment

This summary is informational and does not modify RFCs, tasks, or runtime behavior. It is intended to support governance review by grounding security discussions in a common taxonomy and evidence trail.

Security Policies: Governance and Incident Response

Security Policy

- **Scope:** Applies to Anthesis services (Phloem, Xylem, CLI, plugins), all environments, personnel, and third-party integrations.
- **Principles:** Human sovereignty, defense-in-depth, least privilege, traceability, security by design, fail securely.
- **Core Requirements:** Authn/z for all API access, input validation, secrets management, data protection in transit/at rest, monitoring and alerting, secure SDLC practices, incident response adherence, and explicit third-party risk controls.
- **Roles:** Security team owns policy and incident response; engineering and DevOps implement controls; management approves changes.
- **Review Cadence:** Annual review or after P0/P1 incidents.

Incident Response Policy

- **Severity Levels:**
 - P0: Critical, immediate response (<15 minutes), all hands, executive notification.
 - P1: High, response within 1 hour, security lead + DevOps lead + on-call, management notified.
 - P2: Medium, response within 4 hours, on-call + security reviewer.
 - P3: Low, response within 24 hours, standard on-call.
- **Phases:** Detection & triage, containment, investigation, eradication, recovery, and post-incident review.
- **Key Actions:** Credential rotation, system isolation, evidence preservation, patching, staged recovery, and formal postmortem with owners and due dates.
- **Communication:** War room for P0/P1, regular status updates, decisions logged; coordinate external notices with Legal/Compliance as needed.

Related Anthesis Security Summaries

- Governed SDLC Security Baseline - Short orientation to Anthesis's current governed SDLC security posture.
- CSSLP Analysis Summary - Anthesis-facing summary of the broader CSSLP security assessment and how CSSLP principles apply to the platform.